# Firm Entry Decisions in a City*

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**Abstract**

This paper examines how competitive and agglomerative forces affect small firms in an urban setting. Extending the classic Bresnahan and Reiss (1991) framework to multiple industries within an urban area, I estimate a structural entry model using maximum likelihood on a novel dataset combining firm locations in the city of Jerusalem, GIS processing, and demographic data. Results show persistent within-industry competition effects, whereby similar firms reduce each other's per-capita variable profits, and between-industry agglomeration effects, whereby firms benefit from collocating with those in other industries. Specific cross-industry interactions are heterogeneous: some industry pairs display complementarities, while others exhibit competition. These findings emphasize the complexity of urban market interactions and the importance of accounting for industry specification in urban entry analysis.

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# 1 Introduction

Cities play a central role in modern economic life. Serving as a center of residence, labor, and culture, the metropolitan environment is in the background of decisions taken by many economic players. These decisions include daily consumption and travel choices from a consumer's standpoint, as well as firms' decisions such as pricing and the variety of goods to offer. One major decision faced by a small firm in a big city is entry. Many times, while a large number of firms may contemplate operating in some of the city's markets, not all ultimately choose to do so.

In this work, I examine the economic conditions and factors that affect entry decisions made by firms in the metropolitan environment. Specifically, I focus on the manner in which the existence of nearby businesses and the type of economic activity they take part in affects their mutual ability to exist profitably, alongside their location in the city. These factors capture the effects of both competitive and agglomeration forces, as well as the influence of demographic distribution and geographic characteristics on firm profitability. Accordingly, the main research question is how are the variable profits of small firms in urban environments affected by the entry of additional firms.

This question may, of course, yield different answers in different scenarios. In particular, it may change depending on whether the additional entrant is a "friend" attracting new consumers, offering complementary products, and increasing overall profits, or a mere competitor, a "foe" that cuts the incumbents' profits. This distinction illustrates the tension between competition and agglomeration in location choice. That is, a firm would wish to avoid locating itself near competitors but may benefit from operating in proximity to other non-competing firms. In this study, I examine how these opposing effects jointly influence variable profits.

Another important factor in entry decisions is the industry in which firms operate. The literature suggests that entry can have heterogeneous effects on profitability across different sectors, implying that while some firms require a large number of customers to operate, others may be profitable given a smaller market size. For example, it seems unlikely for a cinema or a private hospital to operate serving only a few hundred customers, whereas such a market may be sufficient to sustain a small grocery store or a barber. This intuition provides grounds to explore how variable profits changes due to entry depend on the firms' industry and location.

From an empirical standpoint, I wish to provide insights regarding the entry effects of competition and agglomeration described above. Gathering such evidence about the considerations and profitability of firms can improve our understanding of the entry dynamics of economic activity in a city and the resulting spatial organization of economic activities in equilibria. Methodologically, the study extends the classic entry framework of Bresnahan and Reiss (1991) to a metropolitan environment, combining it with a novel, multi-industry dataset built from publicly available firm-location information, geographic information systems (GIS) processing, and official demographic and transport data. This approach leverages the simple but informative fact of firm presence to infer profitability conditions, without requiring detailed micro-data on prices, quantities, or consumer choices, and provides a foundation for future work linking structural entry models with rich urban data.

To address these questions, and achieve those goals I introduce a model involving a two-stage entry game, extending a classic framework first presented by Bresnahan and Reiss (1991). This game involve a large number of potential entrants that simultaneously consider market entry. Firms are assumed to be homogeneous in their profit structure, and may operate in a finite set of fixed shopping centers. The game is played simultaneously in multiple industries operating within the city. For empirical estimation I develop a novel database of firm locations in the city of Jerusalem, with census demographics and density-based clustering. By specifying functional forms for firms' profit functions, the model's parameters are estimated using maximum likelihood. These parameters are then used to draw conclusions regarding the firms' profits and entry decisions.

I estimate the model's parameters for ten selected industries in Jerusalem. The results reveal three main patterns. First, within-industry competition exerts persistent negative effects on per-capita variable profits, with the largest impact typically occurring when a second or third competitor enters—mirroring the patterns found by Bresnahan and Reiss (1991) in non-urban markets. Second, several specifications indicate positive cross-industry agglomeration effects, whereby the presence of firms from other industries raises profitability. Finally, an analysis of bilateral industry relations shows that these cross-industry effects are heterogeneous in both magnitude and direction: some industry combinations exhibit complementarity, creating a positive agglomeration effect, while others display competitive interactions that reduce mutual profits. Together, these findings suggest that the profitability impact of nearby economic activity depends on the nature of the industries involved, and that urban entry patterns are shaped by a complex interplay between competition and agglomeration.

The rest of this work proceeds as follows: Section 2 reviews the literature on firm location choice, entry decisions, and agglomeration economies in the contexts of both industry structure and consumer choice. Section 3 presents the model used to estimate and analyze the research questions. It includes a theoretical model based on the aforementioned model by Bresnahan and Reiss, adjusted to a metropolitan setting, and details the approach taken for parameter estimation. Section 4 describes the construction of the unique dataset built for this work. The process combines user-submitted business locations with government population statistics and transport data, and applies density-based clustering algorithms to define economic centers. The section also discusses the selection of industries included in the analysis. Section 5 reports the estimation process with the different specifications results and their interpretation. Section 6 discusses the work and concludes.

# 2 Literature

**Firm Entry and Location Decisions.** The study of firm entry and location choice is a classic economic issue. Both the theoretical and the empirical literature have explored the factors influencing firms' decisions on whether to enter a market and where to operate. Such research examines how market conditions and constraints shape firms' entry strategies.

The early Central Place Theory (CPT) proposed by Christaller (1966) takes a general equilibrium approach, by solely defining an entry threshold in terms of required number of clients, and a good-range for each industry, determining the maximum distance consumers may travel. This simple theory predicts an equilibrium in which firms cluster in central places, with firms from different industries locating together in clusters, while same-industry firms keep distance from each other to acquire enough customers and pass their entry threshold. Christaller's theory is considered historically as a key model, suggesting that 'central places' emerge to serve the demand for goods in surrounding areas (J. Parr, 2017). However, CPT has been criticized for its simplistic assumptions about consumer behavior (J. Parr, 2017), its failure to apply in poly-centric cities or various commercial forms, and its weakness in empirical applicability (Brown, 1993).

Another class of general equilibrium models called rent-accessibility (RA) models was introduced by von Thünen in 1842, and later expanded by many scholars, notably Alonso (1964) and Sinclair (1967). This family of models places greater emphasis on the firms' profit and cost structure. Specifically, they focus on the trade-off firms face between locating in attractive places of high consumer demand and the high rental costs in those places. Using the equilibrium in the rental market for firms, RA theory suggests that groups of industries with similar cost structure and common good features (e.g., retailers, service providers, farms) will locate to form 'sectoral rings' around the trade center of the city (Capello et al., 2013). While these models offer a structured framework for predicting firm locations rooted in industry-level demand and cost analysis, they have also been criticized for oversimplification and for failing to accurately reflect the observed spatial firm distribution.

Other theories have focused more on the locations of firms with respect to each other.

The canonical Hotelling (1929) model presents a simple result in which, despite competing with each other, in a 'linear city' firms tend to cluster and will find themselves locating near each other in equilibrium. Later formulations of the model integrated with game-theoretic modeling allowed many extensions, and introduced a broader range of equilibrium outcomes. The most famous version, 'the circular city' model by Salop (1979), breeds the other-end extreme result in which firms will instead position themselves as far apart as possible. The theoretical contributions by Hotelling, Salop and their successors have laid ground for integrating game-theoretic models into empirical research on firm location choices, as well as product differentiation models. By focusing on individual industries, rather than requiring a general equilibrium framework, this approach formalized the way firms strategically use location to compete, attract customers and maximize profits.

A new branch of literature emerged with a group of papers that developed empirical models of the market structure. This literature that combines theoretical and structural models of market entry, product variety, and product quality with empirical econometric estimation.

Bresnahan and Reiss's 1991 paper (BR91) is a seminal contribution to this field. BR91 have focused on using firms' market presence as information, an indicator of their profitability. By combining a market entry game with a functional specification of firms' profit functions, they estimated how variable profits change as a result of additional market entry. Their dataset of market incumbency was drawn from Yellow Pages directories, and was focused on small service providers (e.g., doctors, tire dealers, plumbers) across 202 isolated U.S. towns. The key finding was the sharp decline in variable profits with the entry of the second and third competitor, alongside the introduction of a new methodological approach.

Building on BR91's shoulders, a vast literature has extended these methodology and findings, addressing additional research questions by incorporating and considering alternative assumptions. Berry and Reiss (2007) adopt a non-parametric identification approach, allowing for more flexible and less restrictive functional forms of the firm's profit and cost structure.

Another significant extension of BR91's model concerns their main assumption of homogeneous firms, which leads to a single entrant number in equilibrium. In order to incorporate heterogeneous firms in an entry model and manage the resulting challenge of multiple equilibria, several approaches were proposed. Berry (1992) introduced the order of entry decisions as a modeling assumption, while others model the equilibrium selection itself. A different approach taken by Seim (2006) inserts asymmetric information, whereas Ciliberto and Tamer (2009) develop a partial identification method to address these issues.

Another extension, most notably explored by Mazzeo (2002) on the market for motels, is the firms' choice of product quality. In this formulation, firms make both entry and quality decisions, with assumptions about entry order and the costs of quality. A notable contribution by Jia (2008) allowed the distinction between large wholesale chains and small retailers, by analyzing the impact of Walmart's entry on market equilibria.

The main contribution of this vast literature lies in its ability to provide robust empirical answers for questions regarding specific markets. Scholars analyze a single market at a time, without a thorough account for interactions between different industries, nevertheless a concept of general equilibrium as given in the classic CLT and RA models. Another aspect of competition mostly neglected in this branch of literature is the spatial considerations of firms when making location choices. Studies mostly do not examine the distance between retailers, location premium, or the demographics of consumers surrounding specific firms within the same market.[1] This limitation is emphasized by Hotelling's strategic framework and subsequent literature, which suggest that location choice may play a crucial role in determining market shares and product differentiation, affecting a firm's competitive strategy.

**Agglomeration Economies.** The concept of agglomeration economies or agglomeration externalities has been suggested to provide explanations for the observed co-location and clustering of economic activities within regions and cities. The benefits of clustering are essentially positive externalities between firms, which affect each other's profits without internalizing these effects in their competitive decisions (J. B. Parr, 2002). These agglomerative effects may occur both between and within industries, and are traditionally classified into three categories: economies of scale, localization economies, and urbanization economies (Hoover, 1937; Glaeser et al., 1992; Rosenthal & Strange, 2004).

*Economies of scale* refer to cost advantages firms achieve from increased production, leading to lower per-unit costs. While not always linked to agglomeration externalities, they often involve localized benefits like increased specialization and savings from bulk purchasing. *Localization economies* stem from industry-specific economies of scale arising from activity concentration and the creation of an 'industrial ecosystem'. Known as MAR externalities (Marshall, 1890; Arrow, 1962; Romer, 1986), they generate positive effects such as a shared labor pool, lower supplier transport costs, shared infrastructure, and knowledge spillovers. Lastly, *urbanization externalities* arise from sectoral diversity rather than specialization. These effects enhance competitiveness and innovation through cross-sector knowledge spillovers, often found in larger cities (Glaeser et al., 1992). These economies arise from shared infrastructure, institutions, and improved access to services like finance and technology.

Contemporary empirical research on agglomeration economies focuses on using data on

[1] Several papers partly consider those aspects e.g. Seim (2006), previously mentioned

the location of firms' plants, branches, or headquarters to better understand how the industrial environment affects location choices (Dubé et al., 2016). This strand of recent urbanization literature aligns with modern economic models, which incorporate consumer utility and demand estimation to examine firm behavior.

Recent contributions include Dubé et al. (2016), who uses a random utility model with distance-based measures to study business establishments in Canada, to find that proximity to urban centers, firm specialization, and establishment size play significant roles in location decisions. These results are consistent with earlier studies, such as Rosenthal and Strange (2001) and Strauss-Kahn and Vives (2009), which emphasize the influence of agglomeration economies and industry clustering on firm spatial location. Additionally, Jofre-Monseny et al. (2014) and Mota and Brandão (2013) highlight the impact of market size and transportational accessibility on entrepreneurial and industrial site selection, reinforcing the broader importance of spatial economic factors in business location decisions.

**Demand-Side Agglomeration Effects.** Another aspect of agglomeration explored in the literature focuses on its effect on local demand rather than on industrial organization at the regional or city level. The main concept considers the travel decisions made by individuals, with the existence of several firms at the same vicinity shifting demand from the perspective of the single industry. This approach lies in the basis of Spatial interaction models (SIMs), a branch that derived from traditional gravity models. These explore the flow of people, goods, or information over space that results from a decision-making process. These models are commonly used in geography and urban sciences, and were first used in economic literature to analyze urban retail (Wrigley, 1994). Economic SIMs primarily focus on modeling the individual consumer's shopping behavior in a metropolitan environment. By aggregating individual consumption choices, with emphasis on the location of consumption, demand is generated for a location, rather than products. The main factors in spatial shopping decisions include travel costs (in terms of time, distance or price), and the expected product prices and variety at the shopping destination. In that sense, the use of SIMs may include the demand-side aggregation effects.

Another group of economic studies on inter-industry effects have focused on tenant composition in shopping malls. The special situation of a mall hosting many firms with central profit-maximizing management, provides ground for research of this inter-firm relations. The mall enables fixing location considerations by individual firms, and puts focus on effects of additional firms on profitability and demand from the mall owner's perspective (Leung et al., 2024). This literature often finds that industry variety shifts demand for the mall, suggesting an agglomerative effect.

All in all, the agglomerative effects on demand were explored in various directions and context. In the context of small firms and urban local demand, the issue was not yet broadly integrated into the main literature of empirical market structure models.

# 3 Model: Entry Decisions in the City

To analyze how entry affects firms' variable profits in a city, I present a model of market entry adapted to an urban economic environment. This section lays out the model in several steps. Section 3.1 follows the framework presented by Bresnahan and Reiss (1991), establishing a similar entry game, while incorporating several adjustments specific to a city's economic environment. I first outline the entry game and derive the key result: a single industry-level solution for the number of operating firms in each location at equilibrium. I then discuss the model's assumptions, their economic implications and justifications, and potential alternative formulations. Section 3.2 outlines the empirical strategy, which applies a maximum likelihood method to estimate the model's parameters.

## 3.1 Entry Game

The model depicts a city that holds $C$ centers of economic activity (interchangeably referred to as *shopping centers*) and $I$ distinct industries that may operate at each center. Both $C$ and $I$ are fixed, natural numbers, implying that new centers or industries do not emerge. A market is defined as a center-industry combination, and the city therefore hosts $C \times I$ unique markets, denoted by $M$. In the context of BR91's work, a center can be thought of as an individual town, each containing $I$ distinct markets— a single one for each industry.

For each market, an infinite number of homogeneous firms consider entry. These firms compete in a two-stage entry game. In the first stage, all firms make simultaneous entry decisions (whether to enter or not). In the second stage, entrants engage in an oligopolistic competition game. The nature of this competition is left unspecified within the model. I denote the number of entrants in market $m$ as $n_m$ or, equivalently, $n_{ci}$ for the market at center $c$ for the products of industry $i$. Additionally, $n_{c,-i}$ denotes the vector of length $I-1$ containing the entrant numbers in all other markets of center $c$.

The profit function for firms in market $m$ is specified as:

$$\Pi_m = \begin{cases} V_m(n_{ci}, n_{c,-i}) - F_m, & \text{entry} \\ 0, & \text{no entry} \end{cases} \tag{1}$$

The term $F_m$ denotes the entry cost in market $m$, identical for all potential entrants to that market. $V_m$ denotes the variable profit of entrants, which depends on the number of local entrants from the same industry, $n_{ci}$, as well as on the vector of entrant numbers in the other $I-1$ local industries, $n_{c,-i}$.

Following the original model, the key assumption is that the variable profit function $V_m$ decreases with $n_{ci}$. Namely, as the number of competitors within a market increases, per-firm variable profit declines. This assumption reflects the notion that the number of firms intensifies the competition among firms, and while this likely leads to lower prices and higher total quantity demanded, the impact on firm-level variable profit is all in all negative.

No additional assumptions are imposed on the direction in which variable profits change with respect to the number of competitors in other industries. This flexibility reflects the notion that some industries may act as net complements or net substitutes to each other. For example, one may think that the entry of a grocery store may reduce the profits of nearby restaurants. This will be the case if consumers consider the two industries as substitutes in a more general 'food market'. Alternatively, the same grocer's entry could benefit neighboring hardware stores, if consumers often wish to purchase goods from those two industries during the same trip, making them net complements. Predicting the nature of these cross-industry relationships would require additional structure on consumer preferences or behavior, which lies beyond the scope of this work. I now proceed to present the main result of the model.

**Theorem 1.** *For each market $m$, given a vector of entry decisions in the other local markets, $\overline{n}_{c,-i}$:*

- *A SPE with $n^*_{ci} = 0$ is received if and only if*

$$V_m(1, \overline{n}_{c,-i}) < F_m$$

- *A SPE with $n^*_{ci} > 0$ is received if and only if*

$$V_m(n^*_{ci} + 1, \overline{n}_{c,-i}) < F_m \leq V_m(n^*_{ci}, \overline{n}_{c,-i})$$

The proof of Theorem 1 is simple, and provided in the appendix. The interpretation of the theorem is intuitive: for any market, if a local monopolist would not be profitable given the entry decisions of all other industries within the same center, then no firms will enter that market in equilibrium. Furthermore, if a positive number of firms $n^*_{ci} > 0$ choose to enter in equilibrium, the result implies that there exists a unique equilibrium number of entrants for that market for every vector $\overline{n}_{c,-i}$.

This result deviates from BR91's, as it may produce multiple possible equilibria at the center level and, by extension, at the city level. This deviation arises due to interactions between local markets within the same center. Specifically, variable profits for a firm depend not only on direct competitors within its own market but also on the entry decisions of firms in other, non-competing markets within the same center. This feature of the model is motivated by the concept of agglomeration introduced previously, purposing that entry in one market can affect the profits of other industries in the same center by shifting local demand. In this sense, an entrant in one market may affect the attractiveness of the entire center, changing the consumer base for other local firms. Such effects would not arise in an isolated-town scenario. If variable profits were instead assumed o be a constant function of $n_{c,-i}$, a unique city-wide equilibrium would exist. In that sense, the model presented here generalizes BR91's model. An example illustrating the potential for multiple equilibria is provided in the appendix.

It is also important to note that the model does not include interactions between firms operating in different centers. In this sense, centers *are* treated as isolated from one another. This standard, simplifying assumption contrasts with some of the literature on consumer behavior, which suggests that consumers choose between multiple centers before deciding what bundle to purchase within their center of choice. If this feature were incorporated, variable profits at one center would be directly affected by entry decisions in other centers across the city. While such interactions may be more realistic, they would also introduce considerable complexity to the model. Although these cross-center profit effects are not explicitly modeled, I account for the spatial relationships between centers in a different way. In the functional specification presented below, variables such as the distance to the nearest competing center and the effective population potentially shared across centers are included. Thus, cross-center effects are implicitly included in the model, through the specified form of the profit function.

As stated previously, existing literature addresses the existence of multiple equilibria by introducing additional structure to the model, either eliminating the multiplicity or structuring the process of equilibrium selection. Addressing this issue is beyond the scope of this work, and thus instead of offering a single equilibrium result that will act as a complete general equilibrium result for the city, I adopt a partial-equilibrium approach. This means that I assume all firms in a market correctly expect the entry decisions in all other local markets, and estimation results are based on the equilibrium that actually materialized. This assumption on expectations can be interpreted of as corresponding to a more realistic dynamic process of entry, in which some incumbent firms already operate from exogenous, historical causes, while others consider entry under the assumption that existing businesses will keep their activity in the future.[2]

## 3.2 Structural Formulation and Estimation Methodology

In order to derive empirical results from the theoretical entry model, additional structure is required. Specifically, the functional forms of the components of the profit function must be specified. This subsection presents the assumed functional forms, describes the maximum likelihood estimation (MLE) strategy used to identify the model parameters, and outlines the type of data required to implement this estimation.

Assume that all $C$ centers of the city are observed. For each market $m$, we observe the number of operating firms $n_m$, and the corresponding industry $i \in \{1, \ldots, I\}$ to which each firm belongs. In addition, we assume the following functional form for the variable profit function:

$$V_m(n_{ci}, n_{c,-i}) = S_{ci} \cdot \pi_{ci}(n_{ci}, n_{c,-i}), \tag{2}$$

where $S_{ci}$ denotes the *market size*, which is the number of customers that visit it regularly or that may visit it. The term $\pi_{ci}$ represents the *per-customer variable profit*. This formulation assumes that total variable profit is proportional to the number of potential customers at the center.

The market size is modeled as a linear function of center-specific characteristics, with industry-specific coefficients:

$$S_{ci} = Y_c \cdot \beta_i, \tag{3}$$

where $Y_c$ is a vector of observable characteristics for center $c$, and $\beta_i$ is a vector of parameters specific to industry $i$. This formulation allows $\beta_i$ to vary across industries, capturing differences between industries at the same center that may, for example, reflect variation in visit frequency or demographic appeal.

The variables in $Y_c$ are center size shifters, features of the center that affect its ability to attract customers. Conceptually, I divide these into three categories. First, population distribution variables, which reflect the center's location premium. These features include economic and social characteristics of the population. Second, transportational attributes, which reflect the center's ease of access for customers from other parts of the city. Third, agglomeration features, which capture the way customers benefit from the presence of diverse economic activities in close proximity. Note that the last type of market size shifters includes

[2] Another important assumption required for the empirical strategy is that centers must not be degenerate at equilibrium, meaning that at least one industry is operational with $n_{ic} > 0$. This is since towns are clearly observed, whereas shopping centers are not, and actual observed firms will be used to define the centers.

variables that reflect inter-industry interactions. Specifically, functions of the entry decisions in other markets, denoted earlier as $n_{c,-i}$.

I model the per-customer variable profit using the following specification:

$$\pi_{ci} = X_{ci} \cdot \alpha_i - \sum_{l=2}^{n_{ci}} \delta_{li}, \tag{4}$$

where $X_{ci}$ is a vector of demand-shifting characteristics for market $(c, i)$, and $\alpha_i$ is a corresponding vector of industry-specific parameters. The term, $X_{ci} \cdot \alpha_i$, represents the per-customer profit of a local monopolist at that market. The second term captures the reduction in per-customer profit due to competition: for each additional entrant beyond the first, a "penalty" $\delta_{li}$ is subtracted. It is assumed that $\delta_{li} > 0$ for all $l \geq 2$, formalizing the assumption that per-firm variable profits decrease with entry. This formulation implies that per-customer profit declines discretely with the number of competitors, and offers flexibility through non-linear entry effects.

Lastly, I assume that while the fixed entry cost $F_{ci}$ is identical for all firms within the same market, it may vary across centers. Specifically, for each industry $i$, these costs are assumed to be independently drawn from a normal distribution:

$$F_{ci} \mid X_{ci}, S_{ci} \sim \mathcal{N}(\gamma_i, \sigma^2), \tag{5}$$

where $\gamma_i$ is an industry-specific mean and $\sigma^2$ is the variance, which is normalized to 1 for identification. The distribution is assumed to be independent and identically distributed (i.i.d.) across centers, conditional on market characteristics.

To estimate the model for a single industry, a maximum likelihood estimation approach is taken. The vector of parameters to be estimated is denoted:

$$\theta_i = (\alpha_i, \beta_i, \gamma_i, \delta_i).$$

The likelihood contribution of center $c$ within industry $i$ is defined as the probability that the equilibrium number of firms, $n^*_{ci}$, equals the observed number of firms $n_{ci}$:

$$\Pr(n^*_{ci} = n_{ci} \mid X).$$

As theorem 1 shows, this probability depends on the equilibrium number of observed firms. Specifically, this probability can be written as the following function $f$, with $X_{ci}$ and $Y_c$ observed in the data:

$$f = \begin{cases} \Pr(F_{ci} > V_{ci}(1, n_{c,-i}) \mid X), & \text{if } n_{ci} = 0, \\ \Pr(V_{ci}(n_{ci}+1, n_{c,-i}) < F_{ci} \leq V_{ci}(n_{ci}, n_{c,-i}) \mid X), & \text{if } n_{ci} > 0. \end{cases} \tag{6}$$

Under the assumption that fixed costs $F_{ci}$ are normally distributed with variance normalized to 1, the likelihood contribution can be written as:

$$f = \begin{cases} 1 - \Phi\left(Y_c \cdot \beta_i \cdot X_{ci} \cdot \alpha_i - \gamma_i\right), & n_{ci} = 0, \\ \Phi\left(Y_c \cdot \beta_i \cdot X_{ci} \cdot \alpha_i - \gamma_i\right) - \Phi\left(Y_c \cdot \beta_i \cdot \left(X_{ci} \cdot \alpha_i - \delta_{2i}\right) - \gamma_i\right), & n_{ci} = 1, \\ \Phi\left(Y_c \cdot \beta_i \cdot \left(X_{ci} \cdot \alpha_i - \sum_{l=2}^{n_{ci}} \delta_{li}\right) - \gamma_i\right) & \\ \quad - \Phi\left(Y_c \cdot \beta_i \cdot \left(X_{ci} \cdot \alpha_i - \sum_{l=2}^{n_{ci}+1} \delta_{li}\right) - \gamma_i\right), & n_{ci} \geq 2. \end{cases} \tag{7}$$

The resulting log-likelihood function over a sample of centers is:

$$\sum_{c=1}^{C} \log \Pr(n_{ci}^* = n_{ci} \mid X). \tag{8}$$

This log-likelihood function is the sum of the natural logarithms of the likelihood contributions $f$ across all observed centers. Since there is no closed-form solution for $\theta_i$, the parameter vector is estimated numerically by maximizing this log-likelihood function.

# 4 Data

This section provides a full description of the data used to estimate and analyze the model. Section 4.1 describes the main source of data —– business locations in Jerusalem obtained from Google's API service. This dataset contains exact geographic coordinates and an economic activity classification for each business. I analyze the business types to identify the industries most suitable for "clean" estimation and discuss the limitations inherent in using a user-fed database. Section 4.2 explains the process of clustering businesses in the city into centers using a density-based unsupervised machine learning method. I detail the HDBSCAN algorithm employed, its relevant features, and its suitability for this economic application, along with various balancing tests used for robustness. This process results in the definition of the city's business centers. Section 4.3 then describes the center-level dataset, which incorporates demographic data from the Israeli Central Bureau of Statistics (CBS) and transportation accessibility measures from the Ministry of Transport (MOT).

Finally, Section 4.4 presents market-level competition measures and motivates the selection of the final subset of industries chosen for estimation and analysis.

## 4.1 Collection of Firm-Level Data

The main data were obtained from Google's "Places API (New)" service via the "Nearby Search" tool. This tool retrieves data collected by Google on notable locations within a specified radius of up to 100 meters from given geographic coordinates (longitude and latitude). Much of this data appears in Google's spatially oriented products and services, such as Google Maps. The information collected on these locations is contributed by users, including mostly visitors, but also business owners or employees. Using geoprocessing tools, the municipal area of Jerusalem was divided into 6,500 circles with a radius of 100 meters, together covering the entire city. The "Nearby Search" tool was then applied iteratively to the centers of these circles, producing a raw database containing all reported locations within the city.

Naturally, user-contributed data processed by a commercial firm such as Google raises some concerns. Namely, it may contain noise or bias, either in certain reported attributes or in the actual presence (or absence) of businesses. A comparison with the municipality's official business documentation used for taxation purposes ("Arnona") suggests that the latter phenomenon is negligible.[3] In addition, several filtering and cleaning procedures, described in the next paragraph, were applied to mitigate these issues.

The original location-level data included 14,339 observations and was filtered in three steps. First, 1,344 observations were removed due to duplication resulting from overlap between the API's search circles. Second, 1,352 businesses were reported as permanently closed. Third, 68 additional businesses were removed as suspected duplicates, sharing identical coordinates and similar names. This last group is likely noise, resulting from the aforementioned origins of observations in visitors' reports. The final filtered dataset contains 11,575 unique observations.

The dataset features each location's name, address, geographic coordinates (longitude and latitude), and whether it is closed. In addition, 74% of locations include an average user rating on a 1–5 scale (using a five-star rating system) as well as the number of ratings received. The remaining 26% of locations have never been rated and thus have no rating score. User rating can be interpreted as a proxy for a firm's product quality, while the

[3] An analysis conducted by the Jerusalem Lab across 61 urban areas defined by the municipality as trade centers found that the Google-based dataset included 9,681 businesses, compared to 10,009 reported by the municipality—a discrepancy of approximately 3%. A comparison across areas yields a mean absolute difference of 40.9 businesses, with a standard deviation of 32.5.

number of ratings may serve as an indicator of customer volume or popularity. However, both measures may be subject to the previously mentioned biases, as they rely heavily on the tastes and demographics of Google's users.

Lastly, each location receives a vector of *types* characterizing the economic activity conducted by the business. There are 62 unique types in total. Some types are general (e.g., `"store"`, `"food"`), while others are more specific (e.g., `"book store"`, `"hair care"`). The types within each vector are ordered from the most specific to the most general. For example, a pharmacy might receive the vector `["pharmacy", "health", "store"]`, while a bakery may be described by `["bakery", "food", "store"]`.

In addition, each type is mapped to one of 13 broader *categories*, each grouping together multiple types. For example, both `"bank"` and `"accountant"` types fall under the *finance* category. One category of particular interest is labeled *general*, which includes five types that may not represent specific industries.

In the context of the model, the type vector determines the industry to which a firm belongs. In the model, each firm is assumed to operate in a single industry and therefore should be assigned only one type for estimation purposes. Figure 1 shows the distribution of the number of types assigned to each location under three different specifications: (i) considering all 62 types, (ii) excluding the 5 that fall under the general category, and (iii) aggregating types by category.

As shown in Figure 1, when considering all types, 51% of businesses are described using a single type, 35% receive two types, 11% receive three types, and the remaining locations (about 3%) are associated with four or more types. When restricting attention to non-general types, 58% of businesses are characterized by a single type, while 34% receive no type. When aggregating types into categories, 68% of businesses are associated with a single category, 27% with two categories, and 5% with three categories.

Having examined the number of types assigned to each business, I now turn to the types themselves. Figure 2 compares the composition of each of the 57 non-general location types. For each type, the figure presents its name (left), the number of locations described with it (right), and three measures indicating how often the type appears alone, which I refer to as "purity": (i) the share of locations for which this is the only type assigned; (ii) the same measure when the five general types are removed; and (iii) the share of locations where this type is always accompanied by at least one other type, even after removing the general types. The figure reveals substantial variation in type frequency: some types, such as `zoo"`, are very rare, while others, like `restaurant"`, are very common. The standard deviation of the number of locations labeled by each type is 196.2. There is also large variation in type purity, with a standard deviation of 41.4 percentage points (39.5 in the

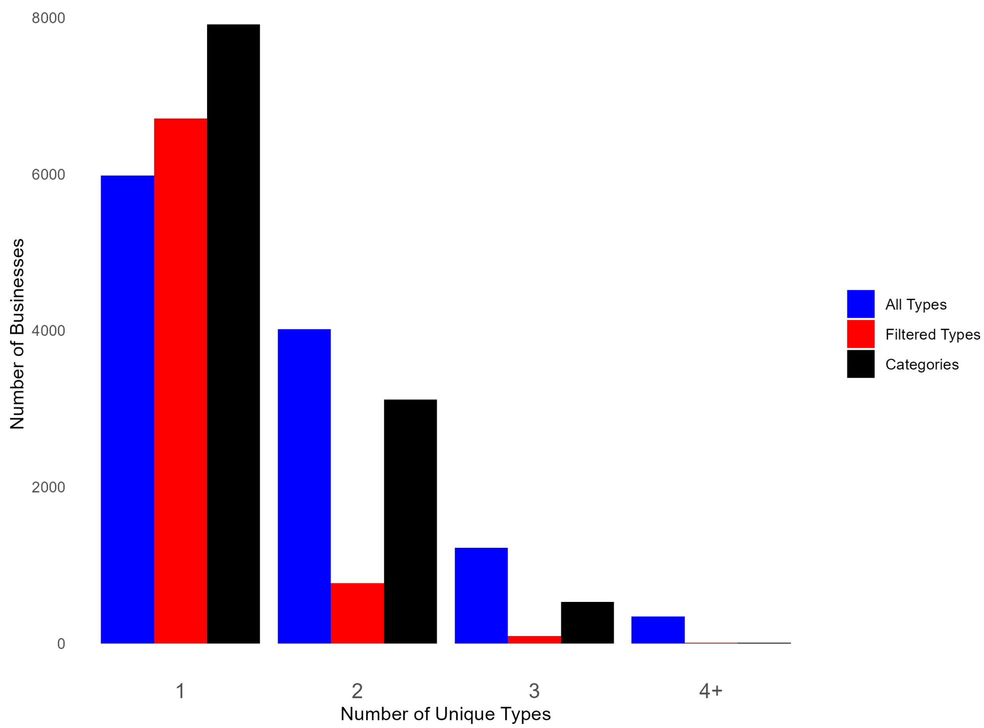



Figure 1: Distribution of Number of Unique Types per Business

Author's calculations based on Google API data. The figure shows the distribution of type vector lengths under three specifications: (1) a baseline including all types; (2) filtered vectors with the general "umbrella" types removed; and (3) alternative vectors using categories that group several types together. By construction, the second and third vectors for each observation are equal to or shorter than the baseline, shifting the distribution to the left.

filtered version). Many types frequently appear as the sole descriptor of a business, while others are almost always assigned together with other types. These characteristics of the data are important: the theoretical framework seeks to analyze clearly defined industries, while the empirical strategy requires sufficient sample size. To address both the multiplicity of industry assignments and the sparsity of certain types, only a selected subset of types will be analyzed using the structural model. The selection process is discussed in a later section.

## 4.2 Clustering to Centers

The concept of business centers as it appears in the model may seem economically intuitive. There are many places in the metropolitan environment that one would easily perceive to be a business center. However, providing a clear and objective division of geographic locations into centers poses a nontrivial challenge. It raises questions such as: How many firms are sufficient to constitute a center? Can an isolated firm be considered a center on its own? How should the boundaries of a center be determined? Must businesses be sufficiently dense or spatially proximate? Should factors such as municipal zoning, neighborhood boundaries, or land-use designations influence the definition?

For the purposes of estimating the model, markets, and thus centers, must be formally defined. In this work, I adopt a statistical approach based solely on the spatial distribution of businesses, imposing minimal assumptions on center characteristics.

Taking such an approach effectively turns the task into a density-based clustering problem, a common technique in the field of unsupervised machine learning. Density-based clustering aims to identify groups of observations that are sufficiently close to one another in feature space to be considered part of the same cluster, while remaining sufficiently distant from other groups to justify separation. When applied to spatial data, the clustering procedure is typically performed using geographic coordinates, sometimes in conjunction with additional features.

Many well-established clustering methods have been developed, each suited to different assumptions or preferences regarding the structure of the resulting clusters. Drawing on prior work on business clustering in urban environments (e.g., Ballantyne et al., 2022), I adopt the HDBSCAN algorithm. This method is well-suited for my task, as it suffices several desirable characteristics.

First, the algorithm avoids relying on any explicit or implicit assumptions about the geometric *shape* of the clusters—a common issue in clustering methods such as the familiar K-means algorithm family. This property is preferred since business centers in cities may take on diverse spatial forms, like forming around a plaza, at the core of a neighborhood, or

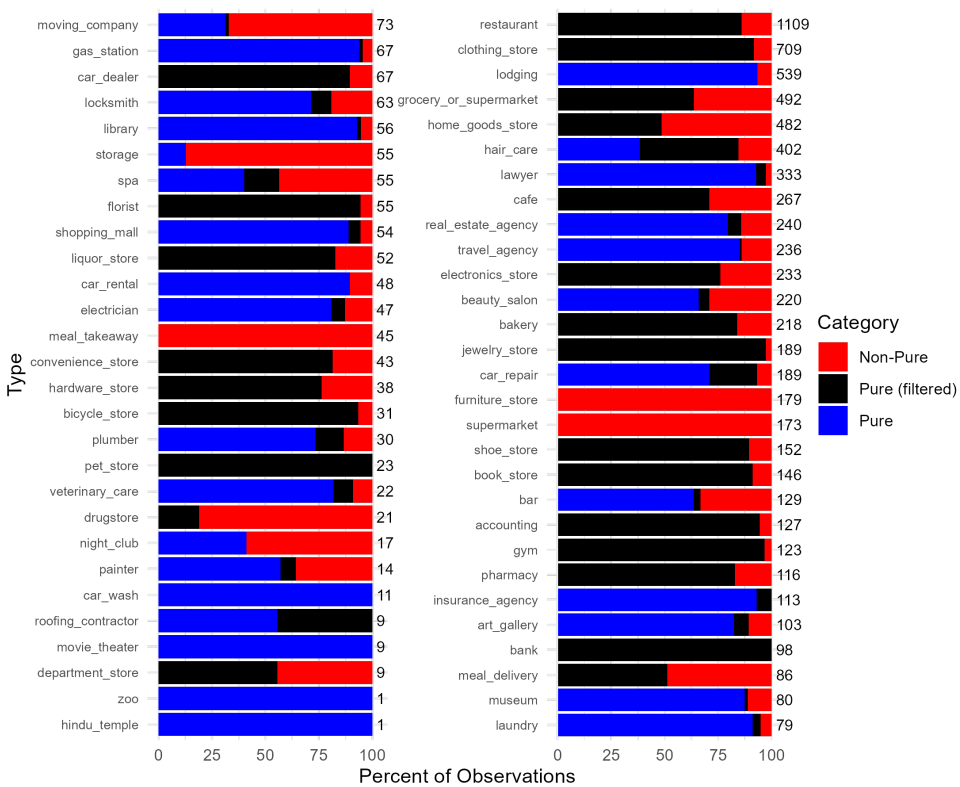



Figure 2: Purity of Firm Types

Author's calculations based on Google API data. The figure presents 57 business types. For each type, the stacked bar shows: (i) the share of locations for which it is the only type assigned, from the total locations described by the type (blue); (ii) the additional share when the five general types are excluded (black); and (iii) the remaining share where it is always accompanied by at least one other type, even after excluding general types (red). The total number of appearances of each type is indicated at the right end of each bar.

along a commercial street. Second, the algorithm does not require the number of clusters to be specified in advance, which is crucial as this number is *a priori* unknown to the economist. Lastly, HDBSCAN allows for the presence of unclustered observations. In other words, not all firms need to belong to a defined center. From an economic perspective, this feature reflects the possibility that some businesses operate in isolation and do not meaningfully belong to any agglomerated cluster. This aspect diverges from the model's structure, which allows entry only at centers. In practice, however, many businesses are spatially isolated, whether due to location constraints such as limited available commercial space, firm-specific limitations, or more idiosyncratic location choices. Such isolated firms are less relevant for the model's entry framework, which emphasizes competitive and agglomerative interactions among nearby businesses.

HDBSCAN[4] is an extension of the traditional DBSCAN algorithm. The original DBSCAN method relies on two tuning parameters: **n**, the minimum number of points required to form a cluster, and **d**, the maximum possible distance between observations that are within the same cluster. In the context of this application, **n** corresponds to the minimum number of firms required to constitute a business center, while **d** reflects the maximum spatial distance between firms considered part of the same center. While the full algorithmic implementation details are beyond the scope of this work, the outcome of DBSCAN is a unique partitioning of the data into clusters that satisfy the given thresholds, with some observations classified as noise.[5] HDBSCAN generalizes this approach by relaxing the requirement to pre-specify **d**. Instead, the algorithm constructs a hierarchy of potential clusters and identifies those that remain consistent over many levels of density. This concept, known as *cluster stability*, captures how persistently a group of observations clusters together across varying thresholds. In effect, HDBSCAN 'automatically' extracts the most stable clusters, which are more likely to reflect genuine economic agglomerations rather than artifacts of arbitrary parameter choices.

It is important to note that the parameters the clustering algorithms receive (**n** and **d**) are not merely technical inputs, but implicitly reflect consumer preferences over firm proximity and density. This means that optimally parameterizing the algorithm requires additional substantive research. This notion illustrates one of the main challenges unsupervised machine learning methods face, which is motivating the calibration of the algorithms with substantial justifications (economic reasoning and empirical research in this case).

It is important to note that the parameters supplied to the clustering algorithm (**n** and **d**) are not merely technical inputs, but implicitly capture consumer preferences regarding the proximity and density of firms. Consequently, choosing these parameters in an economically

[4]Hierarchical Density-Based Spatial Clustering of Applications with Noise.

[5]Noise points are observations that do not meet the density requirements to belong to any cluster.

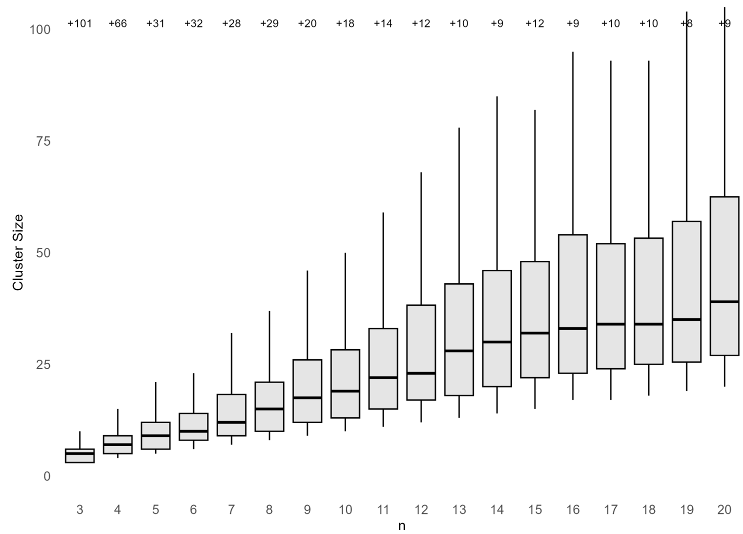
meaningful way requires additional substantive research. This notion highlights a central challenge inherent to unsupervised machine learning methods: parameter choices must be supported by clear justifications, in the current case both economic reasoning and empirical evidence, rather than being set arbitrarily.

I address this challenge in two ways. First, as mentioned, I apply the *hierarchical* DBSCAN algorithm, which relaxes the need to pre-specify the distance parameter **d**. By allowing distances to vary endogenously across clusters, this approach reduces the burden of selecting a seemingly arbitrary city-wide distance threshold. As a result, only one tuning parameter remains: the minimum number of firms required to form a cluster, **n**. This feature partially mitigates the issue I presented, and also allows for more flexible center sizes, which meets the model's prediction given different demand and market size shifters. Second, I run the algorithm iteratively, varying the parameter **n** from 3 to 20. This procedure enables robustness and sensitivity checks with respect to the definition of a center. The range is chosen to encompass two clear intuitive extremes, ensuring that the optimal choice of **n** lies within a plausible, well-motivated interval.

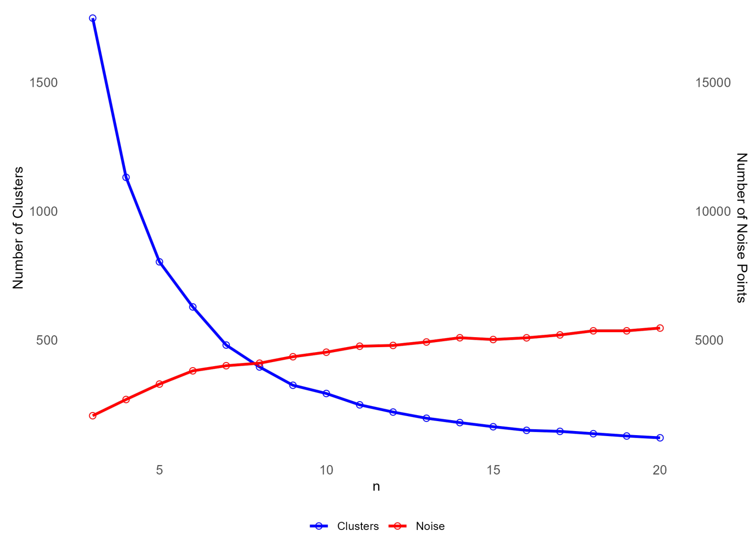
Figure 3 presents descriptive statistics from applying HDBSCAN to the 11,575 observed locations iteratively. Panel (a) shows a box plot of the distribution of cluster sizes for each value of the minimum-points parameter, **n**. The numbers above each box indicate the count of size outlier clusters for that setting. Cluster size statistics increase monotonically with **n**, while the number of outlier clusters drops sharply until $\mathbf{n} = 5$, suggesting that smaller values may produce unstable or over-fragmented clustering. For $\mathbf{n} > 11$, the upper tail of the distribution begins to stabilize, and beyond $\mathbf{n} = 15$ the inter-quartile range (25th–75th percentile) also converges. This pattern indicates that most cluster structures remain consistent beyond this point, with further increases in **n** primarily merging only the largest centers.

Panel (b) plots the total number of clusters (blue) and the number of noise points (red) for each value of **n**.[6] As expected, the number of clusters decreases while the number of noise points increases with higher values of **n**. The primary convergence of clusters occurs by $\mathbf{n} = 10$, after which the rate of change slows considerably. For noise points, three phases emerge: a sharp increase for $\mathbf{n} < 7$, a moderate rise between $\mathbf{n} = 7$ and $\mathbf{n} = 15$, and near stability beyond that. Based on these patterns, the main clustering specification for the analysis sets $\mathbf{n} = 10$. This choice lies just after the main convergence of clusters and preserves a diverse range of cluster sizes. The resulting output includes 292 business centers,

[6] Noise points, as defined by the HDBSCAN method, are firms not allocated to any cluster (shopping center) because they are too far from existing clusters and lack a sufficient number of nearby firms to form a new cluster. The number of noise points therefore depends on the chosen tuning parameters.




(a) Cluster Size Distribution




(b) Number of Clusters and Noise Points

Figure 3: Clustering Results Analysis

Author's calculations. Panel (a) shows the distribution of the number of firms per cluster across specifications of **n** from 3 to 20. Boxes represent the inter-quartile range, and whiskers the 1st–9th decile range. The number of outliers is indicated above each specification. Panel (b) presents the number of unclustered firms (noise) and the total number of clusters under each of the 18 specifications.

encompassing 7,052 firms, which are approximately 61% of all observed locations.

## 4.3 Center-Level Data

Based on the clustering results obtained using HDBSCAN with a chosen minimum cluster size of 10 businesses, I constructed a center-level dataset of 292 business centers, containing a total of 7,052 firms. Each firm was first matched to Central Bureau of Statistics (CBS) data at the statistical area level. The CBS variables include population size, demographic distributions (e.g., age and gender), and socio-economic characteristics such as education, income, and unemployment, which are also summarized into a socio-economic index.

For each firm, I also calculate the population size in surrounding statistical areas by creating three "rings" of increasing area around the firm's original statistical area. Center-level variables are then constructed by averaging the attributes of firms in each cluster. In practice, this means that a center's demographic profile is a weighted average of the statistical areas in the city, with weights determined by the number of firms from each area belonging to the cluster.

Additionally, using geoprocessing of Ministry of Transportation (MOT) data, I add variables capturing bus and road transportation availability. Lastly, measures of cluster size, number of types, and number of categories are included.

Table 1 reports the center-level data summary statistics. It is organized into three groups: demographic characteristics, transportation availability, and center composition. For example, the average business center contains 24 firms and serves an area with an average monthly income per capita of about 4,200 NIS. Centers produce large variation in all variables, suggesting diverse characteristics and underlying market environments across the sample.

## 4.4 Market Analysis and Industry Choice

As noted earlier, the dataset contains a wide variety of business types: some very common, others rare; some typically serving as the sole descriptor of a business, and others frequently appearing alongside additional types. This heterogeneity necessitates a more refined approach to selecting markets that are suitable for the model's analysis. To do this, I define three criteria the type must meet in order to be included in the analysis.

First, the type must have a purity rate[7] exceeding 50%, ensuring that the type is meaningfully represented within its clusters and is commonly sufficient to describe a firm's economic activity. Second, in the main clustering specification ($\mathbf{n} = 10$), the industry must be ac-

[7]Purity rate is defined as the share of times a type appears in the data as the sole definition of firms in their type vector, after eliminating the five general types.

Table 1: Summary Statistics of Selected Center-Level Variables

| Variable | $N$ | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|---|
| **Population Demographics:** | | | | | | |
| Median Age | 287 | 27.18 | 7.08 | 15.00 | 29.57 | 43.00 |
| Monthly Income per Capita (NIS ’000) | 287 | 4.22 | 1.83 | 0.95 | 4.47 | 9.09 |
| Socio-Economic Index | 287 | 4.09 | 2.41 | 1.00 | 4.00 | 9.00 |
| Size–Surrounding (0th Circle, ’000) | 283 | 5.40 | 5.72 | 2.22 | 4.08 | 46.29 |
| Size: 1st Circle (’000) | 283 | 31.86 | 18.74 | 9.48 | 27.64 | 117.89 |
| Size: 2nd Circle (’000) | 283 | 70.60 | 29.64 | 5.70 | 64.62 | 199.87 |
| Size: 3rd Circle (’000) | 283 | 109.93 | 32.76 | 44.02 | 100.32 | 223.86 |
| **Transportation:** | | | | | | |
| Bus Stations (100 m) | 292 | 4.02 | 5.73 | 0.00 | 3.00 | 56.00 |
| Road Segments (100 m) | 292 | 13.50 | 11.89 | 0.00 | 11.00 | 122.00 |
| Road Length (100 m, km) | 292 | 1.50 | 1.74 | 0.00 | 1.01 | 18.55 |
| Bus Stations (300 m) | 292 | 9.42 | 4.79 | 0.00 | 9.00 | 26.00 |
| Road Segments (300 m) | 292 | 33.11 | 13.18 | 4.00 | 30.00 | 78.00 |
| Road Length (300 m, km) | 292 | 4.26 | 1.42 | 0.73 | 4.05 | 9.23 |
| **Center Composition:** | | | | | | |
| Firms in Center | 292 | 24.15 | 17.34 | 10.00 | 19.00 | 128.00 |
| Firm Types in Center | 292 | 13.22 | 5.23 | 1.00 | 12.00 | 39.00 |
| Categories in Center | 292 | 5.34 | 1.44 | 1.00 | 5.00 | 11.00 |

*Note:* Population size variables are measured in thousands (’000); income is reported in thousands of New Israeli Shekels (NIS ’000); and road length is measured in kilometers (km). Demographic variables are missing for a small number of statistical areas, explaining the slightly lower $N$ in the first group of variables. All population variables are matched to firms and averaged for the center. The size circles are calculated as the sum of population in the statistical area (0th circle), its neighboring areas (1st circle), and their neighboring areas up to the 3rd circle, with each area included once. Transportation statistics at the 100 m radius are based on air distance from any firm in the center; at 300 m they are calculated from the geometric center of the cluster’s firms’ coordinates. The minimum observed number of firm types and categories (1) is due to a healthcare center in Jerusalem containing 11 firms, all belonging to the same type and category; the minimum for all other centers is 4 types and 3 categories.

tive in at least 90 clusters to ensure sufficient statistical power. Lastly, using the iterative clustering procedure described above, I compute for each specification between $\mathbf{n} = 3$ and $\mathbf{n} = 15$ the percentage of centers in which the industry operates ($n_{ci} > 0$). For a type to be selected, it must be active in more than 20% but less than 80% of clusters in at least nine of the thirteen iterations. This ensures that selected industries are neither too rare (e.g., `pet store`, `bicycle store`) nor so common and general that they fail to describe a distinctive industry.

Figure 4 illustrates the filtering process. The vertical axis reports the number of clustering specifications in which an industry meets the third criterion (ranging from 0 to 13), and the horizontal axis shows the number of clusters in which the industry is active in the main specification. Industries meeting the prevalence criterion in nine or more specifications are highlighted in black; those satisfying the minimum presence criterion are marked with a diamond. Applying all three conditions, including the first purity requirement, reduces the list of 57 industry types to the following ten: `bakery`, `beauty salon`, `cafe`, `clothing store`, `electronics store`, `contractor`,[8] `grocery or supermarket`, `hair care`, `home goods store`, and `restaurant`.

Figure 5 presents the ten selected industries, showing the number of markets they serve and the distribution of same-industry competitors within those markets. Markets with five or more competitors are aggregated into a single category. For all industries except restaurants, roughly half of markets are inactive ($n_{ci} = 0$), and in the active markets, firms most commonly operate as local monopolies or duopolies. In contrast, the restaurant sector exhibits a wider distribution of competitor counts across centers. This suggests that the estimates for $\delta_\ell$ ($\ell > 2$) in this industry may be associated with smaller standard errors and, consequently, stronger statistical power relative to other sectors.

# 5 Estimation and Results

## 5.1 Competitive Effects

The first empirical analysis examines competitive effects within each industry. For the ten selected industries, I estimate the parameters $\alpha_i$, $\beta_i$, $\gamma_i$, and $\delta_i$ introduced earlier. The estimation sample includes 267 of the 292 centers identified by the density-based clustering. Nine centers are excluded due to missing demographic data. In addition, the largest 16

[8] The `contractor` industry includes different types of hardware and construction stores, and home repairing service suppliers.

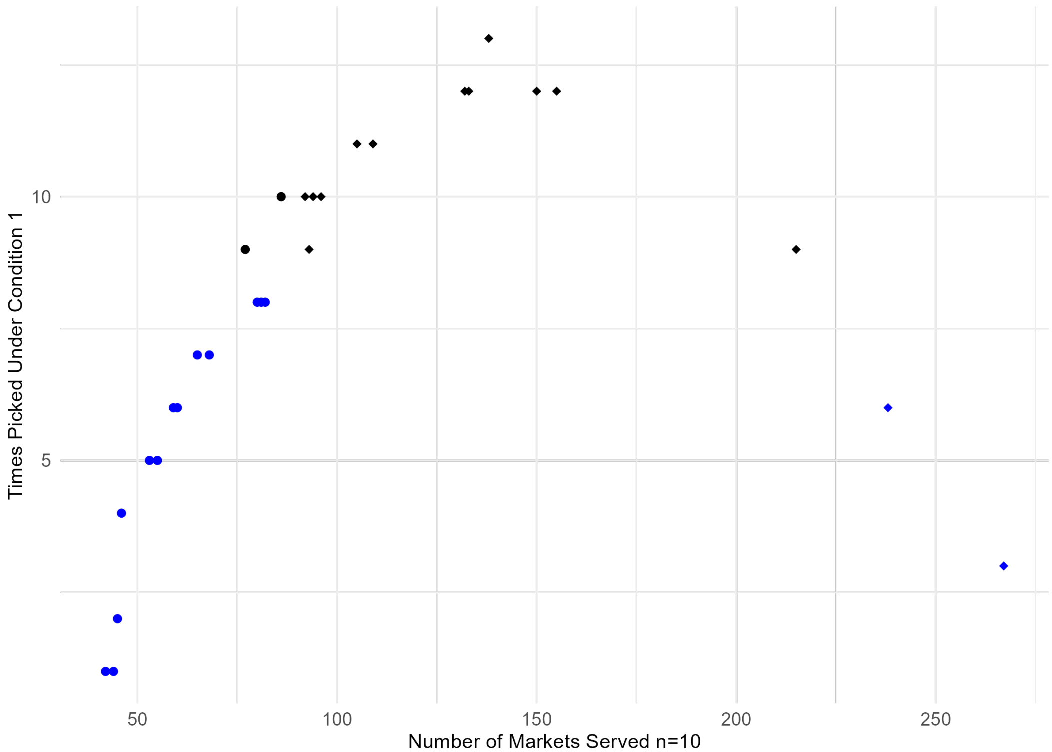



Figure 4: Industry Conditions

Author's calculations. The figure plots industry selection metrics. The horizontal axis reports the number of non-degenerate markets in which the industry is active under the HDBSCAN clustering with the minimum number of firms set to 10. By construction, this axis ranges from 0 to 292, the total number of clusters. The vertical axis shows the number of clustering specifications ($\mathbf{n} = 3$ to $\mathbf{n} = 15$) in which the industry is active in more than 20% but less than 80% of markets, with a maximum possible value of 13.

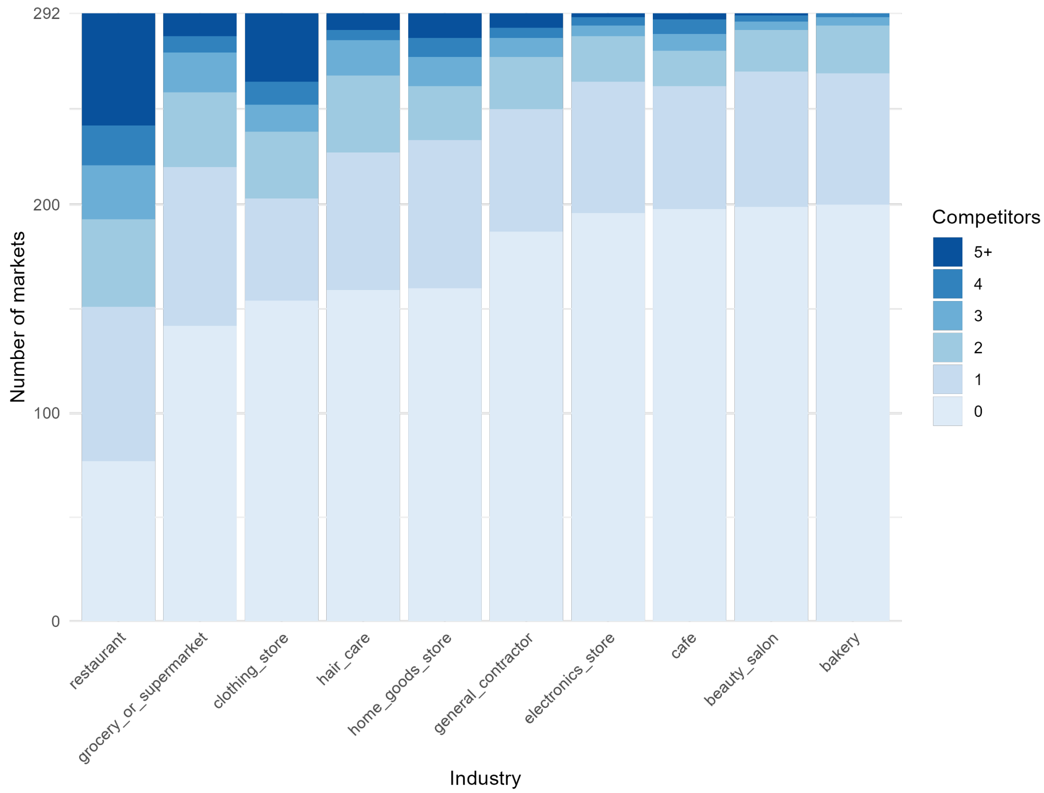



Figure 5: Competitor Number Distribution by Industry

Author's calculations. For each of the 10 selected industry, The figure presents the distribution of the number of competitors at the 292 markets defined by firm clustering. Values higher than 5 were aggregated.

centers are removed manually.[9] These correspond to the top 5% of centers in terms of firm counts, including large enclosed shopping malls (e.g., Malha Mall), wholesale-oriented industrial zones (e.g., Talpiot), and parts of the city center. Such locations are less likely to reflect the entry game modeled, as the influence of surrounding population and local firm entry is likely to be limited relative to smaller, more localized centers.

The baseline specification defines the variable matrices as follows:

$$\begin{aligned} X_{ci} &= \{\text{Income}\}, \\ Y_c &= \{\text{Pop\_0}, \text{Pop\_1}, \text{Pop\_2}, \text{Bus\_Stations}, \text{Road\_Length}\}. \end{aligned}$$

In this setup, per-capita variable profits in a market are shifted by the average monthly income per capita (`Income`). Center size is shifted by the population residing in: (i) the statistical areas containing the center's firms (`Pop_0`), (ii) adjacent areas (`Pop_1`), and (iii) secondary adjacent areas (`Pop_2`). Two transportation-related variables are also included: the number of bus stations within 100 meters of any firm in the center (`Bus_Stations`) and the total length of main roads within the same radius (`Road_Length`). Entry effects are estimated separately for the second through fifth entrants, with all cases where $n_{ic} > 5$ grouped into the fifth entrant category.

This baseline estimation does not incorporate cross-industry agglomeration effects. Instead, it closely follows the functional form and estimation done by BR91, focusing only on within-industry competition. Table 2 reports the estimation results for the ten industries included in the analysis.

Examining the results yields several important observations. First, the estimation of standard deviations for the `electronics store` parameters was unsuccessful: the variance–covariance matrix of the explanatory variables contained values too large, preventing convergence. Second, the competitive effects captured by the $\delta$ parameters replicate the main finding of BR91. For the remaining nine industries, both the effect of the second entrant ($\delta_2$) and the third entrant ($\delta_3$) are statistically significant. The effect of a forth entrant ($\delta_4$) is statistically significant for all but the `bakery` industry, at least at the 10% level, while the effect of entrants from the fifth onward is significant for seven industries (all but `bakery` and `beauty salon`). The absence of significance in these two industries may be driven by a limited number of observations with four or more competitors, as shown earlier in Figure 5.

Another trend seen in the table is a diminishing effect of entry, also appearing in BR91.

[9] All results are not sensitive to the omission of the largest centers, they are excluded from economic and modeling reasoning.

Out of nine successful estimations, the entry estimators of eight present either a consistent downward trend, or mixed trend of constant and downward changes. The exception is only of $\delta_{5,cafe}$, which is much larger than its previous estimates $\delta_{3,cafe}$, $\delta_{4,cafe}$, and similar in size to $\delta_{2,cafe}$. This may suggest either a missing factor that especially affects the coffee shop industry and is not captured in the model, or that cafes revenues act differently than the other examined industries. Lastly, I note that other control variables do not present any consistent patters of direction or significance ($\alpha$ and $\beta$ parameters).

I perform five additional estimations to assess the results' robustness and consistency. First, instead of measuring transportation-related market size shifters within 100 meters of any firm in a center, I alternatively use a 300-meter radius from the geometric center of the cluster. Second, I replace the per-capita income variable with the average socio-economic index score of the surrounding population. Third, I modify the population variables by either excluding the second population "ring" (`Pop_2`) or including an additional "ring" (`Pop_3`). Finally, the fifth specification follows BR91's treatment of fixed costs, which incorporates the effect of entry on fixed costs as well, formally:

$$F_N = \gamma_1 + \sum_{n=2}^{N} \gamma_n, \tag{9}$$

where $N$ denotes the number of entrants.

The first four alternative specifications yield results broadly consistent with the baseline estimates. The only notable change concerns the `electronics store` industry: in these first four robustness checks, standard deviations are successfully estimated, and in the second and fourth specifications, the resulting estimates are statistically significant and similar in magnitude and trend to the baseline estimates for `bakery`.

The final specification incorporates an additional entry effect on fixed costs, which increases the correlation between estimated variables and, in turn, reduces the possibility of successfully estimating standard deviations. As a result, this specification produces many non-estimated standard deviations and parameters estimated at zero due to imposed lower bounds. Nevertheless, the first two $\delta$ parameters remain significant in five industries, and exhibit a diminishing pattern ($\delta_2 > \delta_3$) in four of these cases. Full results for all robustness checks are reported in Appendix C.

## 5.2 Agglomeration Effects

I now consider potential agglomeration effects, shifts in per-capita variable profits for industry $i$ induced by entry in *other* industries within the same center. To allow for such cross-industry influences, I augment the per-customer profit shifters as follows:

Table 2: Estimation Results for Competitive Effects

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -1.113 | -0.992* | -0.271 | -0.043 | -1.472 | -0.555 | -1.489* | -0.743 | -0.204 | 1.280 |
| | (0.704) | (0.526) | (0.355) | (0.036) | (1.026) | ( NA) | (0.805) | (0.598) | (0.437) | (1.324) |
| Pop 0 ($\beta_1$) | 0.004 | -0.088 | -0.149 | 1.528 | -0.014 | -0.039 | -0.054 | -0.020 | 0.024 | 0.060 |
| | (0.064) | (0.070) | (0.193) | (1.011) | (0.040) | ( NA) | (0.034) | (0.089) | (0.182) | (0.080) |
| Pop 1 ($\beta_2$) | -0.037 | 0.012 | 0.122 | -0.281 | 0.007 | -0.057 | 0.004 | 0.021 | -0.080 | 0.022 |
| | (0.024) | (0.018) | (0.170) | (0.199) | (0.011) | ( NA) | (0.010) | (0.021) | (0.204) | (0.031) |
| Pop 2 ($\beta_3$) | 0.034* | 0.017 | -0.018 | 0.211 | 0.009 | 0.060 | 0.010 | -0.001 | 0.055 | 0.002 |
| | (0.020) | (0.012) | (0.034) | (0.157) | (0.008) | ( NA) | (0.007) | (0.008) | (0.135) | (0.006) |
| Bus Stations ($\beta_4$) | -0.004 | -0.015* | -0.011 | -0.138* | -0.007 | -0.008 | -0.009* | -0.008 | -0.014 | 0.011 |
| | (0.006) | (0.009) | (0.028) | (0.074) | (0.006) | ( NA) | (0.005) | (0.007) | (0.037) | (0.015) |
| Road Length ($\beta_5$) | -0.009 | 0.003 | 0.078 | 0.311 | -0.023 | -0.010 | -0.037 | -0.023 | 0.092 | -0.030 |
| | (0.020) | (0.021) | (0.120) | (0.241) | (0.019) | ( NA) | (0.023) | (0.027) | (0.151) | (0.039) |
| $\delta_2$ | 0.897*** | 0.977*** | 0.640*** | 0.247** | 0.754*** | 0.776 | 0.804*** | 0.745*** | 0.596*** | 0.622*** |
| | (0.116) | (0.124) | (0.181) | (0.099) | (0.097) | ( NA) | (0.088) | (0.089) | (0.201) | (0.084) |
| $\delta_3$ | 0.945*** | 0.758*** | 0.368*** | 0.234** | 0.590*** | 0.890 | 0.711*** | 0.701*** | 0.329*** | 0.356*** |
| | (0.232) | (0.185) | (0.125) | (0.092) | (0.119) | ( NA) | (0.113) | (0.111) | (0.119) | (0.064) |
| $\delta_4$ | 0.191 | 0.411* | 0.222** | 0.103** | 0.403*** | 0.412 | 0.789*** | 0.415*** | 0.323** | 0.253*** |
| | (0.187) | (0.231) | (0.107) | (0.048) | (0.137) | ( NA) | (0.183) | (0.122) | (0.126) | (0.057) |
| $\delta_5$ | 0.949 | 0.229 | 0.676** | 0.127** | 0.260* | 0.946 | 0.273** | 0.351** | 0.364** | 0.221*** |
| | (0.908) | (0.224) | (0.307) | (0.059) | (0.147) | ( NA) | (0.133) | (0.153) | (0.164) | (0.057) |
| Log-Likelihood | 213.664 | 223.131 | 251.154 | 352.399 | 269.415 | 220.846 | 315.667 | 319.808 | 319.873 | 443.376 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$). No bounds were used except $\delta \geq 0$.

$$X_{ci} = \{\texttt{Income},\ \texttt{Agglomeration}\},$$

The `Agglomeration` variable is introduced under five alternative definitions, enabling a flexible interpretation of potential agglomerative forces: (1) the number of distinct **categories** active in the center; (2) a binary indicator equal to one if more than five **categories** are active in the center, and zero otherwise; (3) the number of operating **industries** (out of all 62 identified); (4) the number of operating **industries**, restricted to the ten selected industries; and (5) the total number of operating **firms**.

Table 3 reports the results of five different estimations for each industry, each uses a different specification for agglomeration from the ones mentioned above, and presented in a single row. All other parameter estimates omitted, and the full results are provided in the Appendix. The lower part of the table summarizes meta-information regarding the estimation of the four $\delta$ parameters for each industry. Specifically, it presents: (i) the average number of successful estimations; (ii) the average number of statistically significant estimates; and (iii) the number of specifications (out of five) in which the estimated competitive effects exhibit a diminishing pattern for the first two entrants, i.e., $\delta_2 > \delta_3$.

Two main findings emerge from Table 3. First, the competitive effects found in Subsection 5.1 are consistently replicated when considering different agglomeration per-capita profit shifters. Nearly all estimations converge successfully, and the estimated competitive entry effects remain stable in both significance and magnitude relative to the baseline specification without agglomeration. Moreover, the diminishing pattern in the effects of the second and third entrants persists in all industries except `contractor`. Overall, this stability suggests that omitting cross-industry effects in the baseline did not introduce substantial bias into the estimated competitive effects. Second, although not technically bounded, all statistically significant estimates for the agglomeration effect are positive, suggesting that, on average, entry in other industries does not reduce the profitability of firms within a given industry.

I consider differences between the ten examined firms. Five industries present a consistent positive effects (Cafe; Clothing; Electronics; Home Goods; Restaurant) across a large majority of specifications, suggesting they are prone to better profit from other industries on average. The other five industries have a significant effect on a single or two specifications, suggesting a smaller effect on average.

I also find notable heterogeneity across the ten examined industries. Five industries (`cafe`, `clothing`, `electronics`, `home goods`, `restaurant`) exhibit consistently positive and statistically significant agglomeration effects across the majority of specifications, suggesting

Table 3: Estimation Results for Agglomeration Effects

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| # Center Categories ($\alpha_2$) | -0.078 | -0.300 | -0.320 | 0.150*** | 1.148*** | 0.175*** | -0.291 | -1.249 | 0.568* | 0.303** |
| | (0.719) | (0.633) | (1.232) | (0.046) | (0.342) | (0.064) | (0.389) | (1.298) | (0.291) | (0.150) |
| # Center Categories > 5 ($\alpha_2$) | -0.561 | -0.247 | 0.660*** | 0.281** | 0.378 | 0.629*** | -0.281 | -1.555 | 2.110** | 1.345** |
| | (1.162) | (1.387) | (0.240) | (0.130) | (2.254) | (0.239) | (1.221) | ( NA) | (0.991) | (0.685) |
| # Center Industries ($\alpha_2$) | -0.012 | -0.148 | 0.708*** | 0.039 | -0.661 | 0.066* | 0.033*** | -0.409 | 0.185*** | 0.198* |
| | (0.121) | (0.265) | (0.034) | (0.027) | (0.550) | (0.037) | (0.012) | (0.448) | (0.071) | (0.107) |
| # Center Chosen Industries ($\alpha_2$) | 0.590* | 0.423* | 0.202* | 0.184** | 1.722 | 0.195* | 0.079** | 0.911* | 0.189* | 1.243* |
| | (0.338) | (0.243) | (0.121) | (0.077) | (1.084) | (0.101) | (0.040) | (0.491) | (0.108) | (0.654) |
| # Center Firms ($\alpha_2$) | -0.062 | -0.059 | 0.229*** | 0.027* | 0.339 | 0.053*** | -0.170 | -0.369 | 0.185** | 0.146 |
| | (0.100) | (0.080) | (0.071) | (0.016) | (0.247) | (0.015) | ( NA) | (0.424) | (0.093) | (0.537) |
| # Successful | 4 | 4 | 4 | 4 | 3.8 | 4 | 4 | 3.8 | 4 | 4 |
| # Significant | 2 | 3 | 4 | 3.6 | 2 | 4 | 4 | 3.8 | 4 | 4 |
| # $\delta$ Diminish | 4 | 5 | 5 | 5 | 0 | 5 | 5 | 5 | 5 | 5 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. The upper panel reports five distinct estimations for each industry, using a different agglomeration variable: # Center Categories, # Center Categories > 5, # Center Industries, # Center Chosen Industries, # Center Firms — alternative agglomeration measures as defined in Section 4. All variables enter the per-capita variable profit function linearly via $\alpha_2$ coefficients. Summary rows: # Successful — average number of converged standard deviations for $\delta$ parameters; # Significant — average number of significant $\delta$ coefficients; # $\delta$ Diminish — number of cases where $\delta_2 > \delta_3$.

that they tend to benefit more from the presence of other industries. In contrast, the remaining five industries display statistically significant agglomeration effects in one or two specifications, indicating a smaller and less systematic average effect.

The five agglomeration specifications capture related but different aspects of this notion. The first two focus solely on the diversity of economic activity within a center, measured either by the number of distinct **categories** (which aggregate multiple types/industries) or by a binary indicator equal to one if more than five categories are active. The latter captures the possibility that agglomeration benefits may arise only in highly-enough diverse centers, allowing for a non-linear effect. The third specification measures the total number of operating **industries** out of all 62 identified, providing a finer breakdown than categories while retaining full industry coverage. The fourth narrows this count to the ten selected industries, reflecting the idea that other industries' labeling is less indicative, so their presence may not reflect consistent economic activity. In such cases, the economic impact on the ten selected industries may vary widely, potentially diluting any average effect. This restriction improves interpretability but reduces statistical power by considering only 10 of 62 possible industries. Finally, the fifth specification uses the total number of operating **firms** in the center, which reflects the overall level of business activity regardless of its composition. Overall, results are broadly similar across the five specifications, suggesting that each captures an overlapping component of the agglomeration effect.

I note that the specifications in Table 3 assume that the agglomeration effect is linear in the chosen variable, although the second specification does use a dummy variable. I now consider an alternative, more flexible specification to estimate the agglomeration effects:

$$\pi_{ci} = X_{ci}\alpha_i - \sum_{l=2}^{n_{ci}} \delta_{li} + \sum_{k=1}^{n_{c,-i}} \epsilon_{k,-i}, \tag{10}$$

where $X_{ci}$ contains only `Income`, and $Y_c$ is left unchanged. Here, the new $\epsilon$ parameters measure the change in $\pi_{ci}$ associated with the $k$-th entrant from a *different* industry. This formulation allows the agglomeration effect to vary non-linearly with the number of other-industry entrants, rather than being restricted to a single linear coefficient as before. In this implementation, $n_{c,-i}$ is defined as the number of distinct **categories** active in the center, differently than in the original model. I again limit the parameters at five entrants, treating $n_{c,-i} > 5$ as five. Table 4 reports the estimation results for this version.

The estimation results reveal two main patterns. First, competitive effects largely persist even with the inclusion of five additional parameters. Second, the non-linear agglomeration effects are mostly statistically insignificant. The main exceptions occur for the entry of the first category, $\epsilon_1$, and for $\epsilon_5$, which captures the average effect of five or more categories.

Table 4: Estimation Results for Non-linear Agglomeration Effects

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -0.027 | 0.041 | 0.054 | -0.103 | 0.027*** | -0.039*** | 0.025 | 0.076 | -0.074 | -0.015 |
| | ( NA) | ( NA) | (0.051) | (0.103) | (0.001) | (0.010) | (0.019) | ( NA) | ( NA) | (0.023) |
| Pop 0 ($\beta_1$) | -1.348 | -0.671 | -0.713 | 0.736 | -0.301 | 0.255 | -0.410 | -0.631 | 0.467 | -0.622 |
| | ( NA) | ( NA) | (0.593) | (0.840) | (0.356) | (0.757) | ( NA) | ( NA) | (0.313) | (1.020) |
| Pop 1 ($\beta_2$) | -0.043 | -0.117 | 0.352 | -0.177 | 0.177 | -1.070 | 0.028 | -0.052 | -0.480 | 0.219 |
| | (0.269) | (0.151) | (0.369) | (0.194) | (0.161) | ( NA) | (0.075) | (0.166) | ( NA) | (0.361) |
| Pop 2 ($\beta_3$) | 0.530 | 0.380 | 0.048 | 0.207 | 0.266 | 1.100 | 0.346 | 0.316 | 0.424 | 0.221 |
| | ( NA) | ( NA) | (0.087) | (0.203) | ( NA) | ( NA) | ( NA) | ( NA) | ( NA) | (0.278) |
| Bus Stations ($\beta_4$) | 0.035 | -0.039 | -0.064 | -0.088 | -0.062 | -0.028 | -0.079*** | -0.072 | -0.073*** | 0.084 |
| | (0.078) | (0.027) | (0.067) | (0.086) | ( NA) | (0.098) | (0.024) | ( NA) | (0.028) | (0.115) |
| Road Length ($\beta_5$) | 0.225 | 0.167** | 0.737 | 0.388 | 0.178 | 0.051 | -0.047 | 0.384 | 0.423 | 0.118 |
| | ( NA) | (0.085) | (0.594) | (0.418) | (0.140) | (0.336) | (0.080) | ( NA) | ( NA) | (0.315) |
| $\delta_2$ | 0.283* | 0.456* | 0.406* | 0.376** | 0.278 | 0.206*** | 0.368*** | 0.320** | 0.405** | 0.313** |
| | (0.162) | (0.247) | (0.222) | (0.159) | ( NA) | (0.030) | (0.053) | (0.157) | (0.192) | (0.133) |
| $\delta_3$ | 0.300* | 0.386* | 0.247** | 0.369** | 0.201 | 0.299*** | 0.348*** | 0.304*** | 0.223*** | 0.151** |
| | (0.163) | (0.207) | (0.107) | (0.164) | ( NA) | (0.075) | (0.066) | (0.107) | (0.074) | (0.074) |
| $\delta_4$ | 0.052 | 0.179 | 0.167* | 0.168 | 0.127 | 0.197 | 0.409*** | 0.182** | 0.229*** | 0.101*** |
| | (0.075) | ( NA) | (0.090) | (0.105) | ( NA) | (0.167) | (0.107) | (0.075) | (0.068) | (0.031) |
| $\delta_5$ | 1.454 | 0.131 | 0.569* | 0.199* | 0.084 | 1.569 | 0.179** | 0.147*** | 0.278 | 0.086*** |
| | (1774.681) | (0.092) | (0.300) | (0.106) | ( NA) | (3.032) | (0.089) | (0.023) | (1.627) | (0.022) |
| $\epsilon_1$ | -0.855 | -0.753 | -1.396 | 0.912* | -0.807 | 1.826** | -0.562 | -0.820* | -0.630 | -0.587 |
| | (1.249) | (0.575) | (1.666) | (0.525) | (0.651) | (0.874) | (0.538) | (0.485) | (1.017) | (0.750) |
| $\epsilon_2$ | 0.754** | -0.102 | 0.377 | -0.640 | 0.351 | 0.871 | 0.249 | 0.125 | 0.674 | 0.430 |
| | (0.364) | (0.606) | (1.456) | (0.894) | (0.765) | (1.875) | (0.551) | (0.626) | (1.022) | (0.684) |
| $\epsilon_3$ | 0.181 | 0.496 | 0.343 | 0.073 | 0.125 | 0.000 | 0.202 | 0.237 | 0.159 | 0.489 |
| | ( NA) | ( NA) | (0.377) | (0.289) | (0.141) | (0.082) | (0.157) | ( NA) | (0.168) | (0.665) |
| $\epsilon_4$ | -0.127 | -0.024 | 0.090 | 0.030 | -0.047 | 0.005 | -0.115 | 0.212 | -0.002 | -0.001 |
| | ( NA) | (0.099) | (0.168) | (0.156) | (0.088) | (0.053) | (0.095) | ( NA) | (0.112) | (0.075) |
| $\epsilon_5$ | 0.023 | -0.021 | 0.288 | 0.338*** | 0.246 | 0.057** | 0.149** | -0.145 | 0.850*** | 0.859*** |
| | (0.017) | (0.077) | (0.260) | (0.128) | ( NA) | (0.029) | (0.073) | ( NA) | (0.303) | (0.207) |
| Log-Likelihood | 217.632 | 224.569 | 233.692 | 351.051 | 284.604 | 223.185 | 337.050 | 322.474 | 321.267 | 442.967 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `Pop_0`, `Pop_1`, `Pop_2` — populations in own/adjacent/secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100m; `Road_Length` — total length of main roads within 100m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$). $\epsilon_k$ — discrete bonuses to per-capita variable profits for the $k$-th operating category ($n_{c,-i} > 5$ grouped into $\epsilon_5$). No bounds were used except $\delta \geq 0$.

These latter parameters are positive and significant in half of the industries. No other $\epsilon$ parameters emerge consistently with statistical significance.

The absence of evidence for discretely incremental agglomeration effects at the category level may stem from limited statistical power. Alternatively, it is plausible that the estimates reflect the way categories aggregate multiple industries, combining heterogeneous effects—some positive, some negligible, and some even negative—which dilutes the average impact. It may also just be possible that agglomeration comes in play mostly or only with large variety of economic activity within centers. This interpretation aligns with the effects found for the fifth and later entrants, and nearly none for the first few entrants.

Both the linear and non-linear specifications treat agglomeration effects in an aggregate manner, without distinguishing between the identities of the categories or industries driving the effects. The next section examines the more specific relationships between particular industries.

## 5.3 Inter-Industry Agglomeration Effects

I consider the bilateral agglomeration effects between the selected industries. Specifically, I specify the model with per-customer profit for industry $i$ is:

$$\pi_{ci} = X_{ci}\,\alpha_i \;-\; \sum_{\ell=2}^{n_{ci}} \delta_{\ell i} \;+\; \sum_{j \neq i} \mathbb{1}\{\, n_{cj} > 0 \,\}\, \zeta_{ij}, \tag{11}$$

Here, per-capita variable profits are shifted by the variables in $X_{ci}$, the within-industry entry effects, and the entry effects from each other industry. The parameter $\zeta_{ij}$ measures the impact of the any positive entrant number from industry $j$ on the per-capita variable profits of firms in industry $i$.

This formulation allows the relationship between two economic activities to vary by industry pair, rather than assuming a homogeneous cross-industry effect. A positive estimate of $\zeta_{ij}$ can be interpreted as an agglomerative advantage, while a negative value may indicate that the industries actually compete, either because consumers view them as substitutes, or because they serve different distinct "types" of consumers.

I estimate the model's parameters with $X_{ci}$, $Y_c$ unchanged. Table 5 presents the resulting matrix of cross-industry agglomeration effects. The estimate for $\zeta_{ij}$ is in row of industry $i$, and the column of industry $j$. Cell colors indicate the the sign and statistical significance of each estimate: blue for positive estimates, red for negative ones, darker hues for lower p-value.

Although a larger share of estimations fail to converge in this specification, several patterns do emerge. Overall, the estimated effects are diverse: some industry pairs appear to compete, reducing each other's profits with $\zeta_{ij} < 0$, while others may be thought of as complements, supporting mutual profitability with $\zeta_{ij} > 0$. A clear example of apparent competition is the mutual negative effect between `restaurant` and `cafe`, where entry in either reduces the other's profits. Conversely, `electronics store` and `contractor` exhibit positive reciprocal effects, suggesting complementary behavior. Some relationships are asymmetric: for instance, `home goods store` entry increases `electronics store` profits, but the reverse effect is not statistically significant. Lastly, although the last couple of examples seem intuitive, not all relations found can be easily explained, and the main takeaway is that inter-industry effect may vary, which called for further research on those relationships, perhaps with larger focus on industry definitions, which can better reveal bilateral agglomeration and competition between firms.

# 6 Discussion & Concluding Remarks

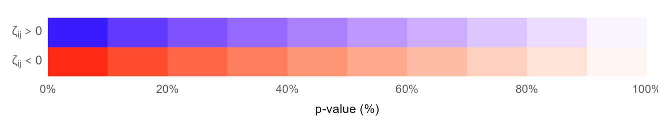
Examining the overall results, methodology, and data used, several issues arise that merit discussion to assess the validity of the process and motivate directions for future research. In addressing the central question of how firms determine their mutual entry decisions in an urban setting, the chosen framework offers clear advantages but also imposes limitations. The Bresnahan and Reiss (1991) model relies on the simple fact of firm presence, with assumed profitability for entry inference. By treating shopping centers as analogues of towns, I incorporate the necessary inter-industry effects on profits. These interactions, however, may generate multiple equilibria, restricting the analysis to partial equilibrium. This non-uniqueness limits meaningful counterfactual analysis based directly on the estimated results. Moreover, the assumption of a large pool of potential entrants and free entry may be reasonable when considering towns, and not taking into account spatial locations. For urban centers, it may be that not every profitable firm eventually enters the market. That is since cities (including Jerusalem) regulate commercial use of real estate, which limits firms' fully flexible entry. Such regulations may divert entrants to peripheral locations or prevent entry altogether. If so, the estimated competitive effects ($\delta$s) may be biased upward in absolute value, as non-entry is attributed to competition rather than regulation constraints.

Another key assumption is firm homogeneity. In reality, some industries exhibit substantial product differentiation, whether vertical or horizontal. This may mean that profit structures are not the same for all firms in an industry, specifically the issue will be more prominent if the differentiation is correlated with demographic variables. For instance, if higher-income areas feature firms that offer higher service quality and facing higher costs, which the model does not capture. The problem may be less severe with horizontal differ-

Table 5: Cross-Industry Effects

| Industry | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Bakery | | -0.090 (0.645) | 0.087 (0.621) | -0.040 (0.286) | -0.008 (0.093) | -0.008 (0.092) | 0.152 (1.079) | 0.110 (0.779) | 0.003 (0.070) | 0.041 (0.304) |
| Beauty Salon | -0.141 (NA) | | -0.019 (NA) | 0.071 (NA) | 0.187 (NA) | 0.192 (NA) | -0.058 (NA) | 0.358 (NA) | -0.264 (NA) | -0.204 (NA) |
| Cafe | 0.318 (0.264) | -0.034 (0.467) | | -0.050 (0.670) | 0.021 (0.304) | -0.164 (2.185) | 0.207 (2.749) | 0.284 (0.252) | -0.197 (2.618) | -0.441* (0.251) |
| Clothing | 0.029 (NA) | 0.139 (NA) | 0.035 (NA) | | -0.218 (NA) | 0.072 (NA) | -0.083 (NA) | 0.238 (NA) | 0.210 (NA) | 0.269 (NA) |
| Contractor | -0.174 (0.108) | 0.176* (0.102) | -0.097 (0.101) | -0.254*** (0.098) | | 0.050 (0.096) | 0.058 (0.089) | 0.011 (0.097) | 0.126 (0.092) | -0.193* (0.099) |
| Electronics | -0.079 (0.073) | 0.082 (0.071) | -0.039 (0.071) | 0.034 (0.060) | 0.061 (0.067) | | 0.011 (0.065) | -0.092 (0.070) | 0.238*** (0.064) | 0.036 (0.065) |
| Grocery | 0.178 (7.360) | -0.022 (0.910) | 0.214 (8.865) | -0.059 (2.465) | 0.105 (4.367) | -0.024 (1.004) | | 0.196 (8.095) | -0.021 (0.876) | 0.024 (1.013) |
| Hair Care | 0.200 (NA) | 0.341 (NA) | 0.087 (NA) | 0.023 (NA) | 0.009 (0.072) | -0.286 (NA) | 0.188 (NA) | | -0.220 (NA) | -0.125 (NA) |
| Home Goods | 0.160 (NA) | -0.313 (NA) | 0.057 (NA) | 0.223 (NA) | 0.261 (NA) | 0.453 (NA) | -0.137 (NA) | -0.041 (NA) | | -0.039 (NA) |
| Restaurant | 0.169 (0.848) | 0.086 (0.139) | -0.306* (0.178) | 0.057 (NA) | -0.071 (0.058) | 0.054 (0.063) | 0.237 (0.318) | 0.013 (0.064) | -0.007 (0.073) | |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Zetas were estimated for each industry; the value in each cell is the $\zeta$ estimated for the industry in the *row* representing the effect of the industry in the *column* on per-capita variable profits. Coloring: darker indicates a lower $p$-value; white indicates NA (standard error not calculated due to non-convergent Hessian); red indicates a negative effect; blue indicates a positive effect. Full p-value coloring scheme:



entiation, which may be captured with the 'type'-based industry definition. That is, while horizontal differentiation may be partly addressed by the industry definitions used, vertical differentiation challenges the assumption of identical profit structures. This concern is partly mitigated by the analysis of ten industries, which generally display similar and stable patterns, suggesting that the main findings regarding competition and agglomeration remain valid.

A further source of heterogeneity stems from spatial variation within centers. After clustering, all firms in a center are assumed to face the same market size. While this may be true in compact centers, with close by firms, businesses at the periphery of more dispersed centers (e.g. a small street near the main gathering of firms) may be exposed to a different market size than its same-center competitors. This issue is another aspect off less flexible entry, due to regulation or the a-priori building in the city faced by the small firm. It is mitigated by excluding the largest 5% of centers in firm number, which are also spatially larger, and by testing robustness across spatial clustering specifications.

Another source of heterogeneity among firms it the questionable notion of city businesses as 'small', while firms may operate under different ownership structure. Although the analysis focuses on industries with broad representation in centers and avoids niche activities or city-wise concentrated markets (e.g. the only three cinemas or the single zoo in Jerusalem), not all firms are "small" in the sense of being independent businesses. Some belong to franchises or large chains, which may have different profit (and specifically cost) structures than small private businesses. Two conditions would preserve the model's validity two conditions are required: (1) all entrants operate under a large chain or none do, meaning that industry firms are indeed homogeneous in that context, and (2) the branch is opened under free entry. Condition (1) may hold, for example, in the case of the clothing shop industry which features mostly branded shops, or in the beauty salon industry that feature small non-franchise businesses. It does not hold for restaurants, for example, which include chains *and* privately-owned businesses.[10] Given a franchised store, the likelihood of condition (2) will depend on the franchise's entry method, it will not hold in the case of large chains that manage branch entry in more complex methods than mere branch-level profitability, and account for branding, minimum branch profits, chain city-wide spreading etc'. To account for this heterogeneity, I rely on analysis across multiple industries and specifications, showing that the results remain consistent. Nonetheless, a fuller treatment of ownership structure would require explicitly incorporating chain affiliation into the profit function.

Turning to the data and processing methods, the novel features of the dataset provide

[10]These claims are based on simple language processing of location names to identify repeated strings, not a full ownership status analysis, which is absent in the data.

both strengths and limitations. The primary data source is user-fed reviews, with their associated advantages and drawbacks already noted. Several pre-described processing steps were taken to adapt these data to the model. First, HDBSCAN clustering was used to define shopping centers, which are not observed opposed to whole towns. While this unsupervised machine-learning method provides robustness and objectivity, it classifies many observations as "noise", effectively excluding non-clustered firms. These firms are treated as non-existent or as exerting only a homogeneous effect. Under the scope of this work, no further steps were taken to include the unclustered firms' effects which may neglect important sources of variation in demand. Within clustered centers, demographic and transportation data were matched using statistical areas. Since many centers overlap multiple areas, weighted averages were used. This approach is reasonable but would be improved by household-level or finer-area data that better reflect the actual customer base and demand structure. Like centers, industry classification was also derived from the data, using Google's typology combined with type-purity calculations. The advantage of this approach lays in the trust in Google's will to provide consumer-useful typology that represents diverse and specific enough economic activities. The limitation is that they may not align perfectly with economic definitions of industries or services. As economists we may wish to better account for the specific services of products sold at the business level to verify differentiability and industry typology.

Finally, statistical power poses another constraint. The number of centers generated by clustering is large enough to allow estimation of the baseline model and some other specifications, but not sufficient to explore extensively richer functional forms or more complex interactions. In addition, the absence of some key variables such as land values or wage data, further restricts the depth of analysis. While the methodology and dataset jointly provide a strong and novel basis for examining urban entry decisions, they also set boundaries on the scope of inference.

To conclude, overall this study demonstrates both the strengths and limitations of the chosen approach. On the positive side, the BR91 framework chosen is simple and tractable, and was used on widely available data on firm locations and demographics, to yields consistent results on competitive and agglomerative effects across multiple industries. By adapting the Bresnahan and Reiss (1991) model to the urban setting and incorporating spatial analysis, the work provides a novel and necessary extension that captures key features of local markets. At the same time, the model's structure and estimation method provide estimates not directly interpretable in terms of simple units of measures like prices, for example, often required for policy analysis or prediction. Moreover, while the analysis covers a broad set of industries, it cannot fully account for firm heterogeneity or product differentiation, and the available data limits a deeper structural analysis.

From a policy perspective, the results should not be interpreted as precise quantitative predictions of entry decisions. Instead, they highlight important qualitative features of local shopping centers' competition and agglomeration. These patterns can inform urban economic policy discussions, particularly in the context of neighborhood centers. For example, if a center is dominated by a single firm in an industry (e.g. a local groceries monopoly), the results suggests that another firm of that same industry may not find entry profitable. Encouraging a more diverse mix of industries, however, could alleviate market power in the monopolized industry by fostering entry, thereby enhancing consumer welfare. While the analysis does not provide direct measures of consumer surplus or pricing outcomes, it sheds light on the mechanisms through which urban retail composition forms.

Future research based on the same framework should build on these findings by employing more refined definitions of both centers and industries, and better account for non-clustered firms and product differentiation.

In order to thoroughly inspect the spatial considerations of consumers, future research will require more detailed demand estimation, explicitly incorporating geographic location alongside product characteristics. Access to richer micro-data including prices, quantities,consumer demographics, and transportation patterns would allow for more precise measurement of demand and supply interactions. Such extensions would enable a deeper understanding of how spatial competition and agglomeration jointly shape urban markets, providing both more accurate insights and more actionable policy implications.

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# Appendix

## A Proof of Theorem 1

I will now lay the full proof of the theorem:

For each market $m$, given a vector local markets' entry decisions $n_{c,-i}$:

- A SPE with $n^*_{ci} = 0$ is received iff $V_m(1, n_{c,-i}) < F$

  $\Rightarrow$ Assume that for a certain market $m$ there exists a SPE such that $n^*_{ci} = 0$. Also assume by contradiction that $V_m(1, n_{c,-i}) \geq F$. Thus, a monopolist may be profitable (or enter at 0 profit). Thus, we receive a contradiction to the fact that this is a SPE, since every firm in the market has a profitable deviation to enter as a monopolist. Thus, $V_m(1, n_{c,-i}) < F$.

  $\Leftarrow$ Assume that for a certain market $m$, $V_m(1, n_{c,-i}) < F$. Also, assume by contradiction that there exists a SPE in which $n^*_{ci} > 0$. Since $V_m(n_{ci}, n_{c,-i})$ is decreasing with $n_{ci}$, it is easy to see that $V_m(n^*_{ci}, n_{c,-i}) < F$. Thus, all entrants in the market have a profitable deviation not to enter, which contradicts the SPE. Thus, $n^*_{ci} = 0$.

  I conclude from this part of the proof that in this game, a SPE with $n^*_{ci} = 0$ is received iff $V_m(1, n_{c,-i}) < F$.

- A SPE with $n^*_{ci} > 0$ is received iff $V_m(n^*_{ci} + 1, n_{c,-i}) < F \leq V_m(n^*_{ci}, n_{c,-i})$

  $\Rightarrow$ Assume that for a certain market $m$ there exists a SPE such that $n^*_{ci} > 0$. Also, assume by contradiction that either $V_m(n^*_{ci}+1, n_{c,-i}) \geq F$ or $F > V_m(n^*_{ci}, n_{c,-i})$. In the first case, there is "space" for another firm to enter the market. This contradicts the SPE, as all non-entrants have a profitable deviation to enter. In the other case, there is not enough "space" for the $n^*_{ci}$ entrants, which all have a profitable deviation not to enter. Thus - a contradiction is received to the SPE, and we have that if $n^*_{ci} > 0$ then $V_m(n^*_{ci} + 1, n_{c,-i}) < F \leq V_m(n^*_{ci}, n_{c,-i})$.

  $\Leftarrow$ Assume that for a certain market $m$, $V_m(n^*_{ci} + 1, n_{c,-i}) < F \leq V_m(n^*_{ci}, n_{c,-i})$. Also, assume by contradiction that there exists a SPE in which $\tilde{n_{ci}}$ firms enter the market, where $\tilde{n_{ci}} \neq n^*_{ci}$. W.L.O.G assume $\tilde{n_{ci}} > n^*_{ci}$. From the assumption that $V_m$ decreases with the number of entrants, we have that since $\tilde{n_{ci}} \geq n^*_{ci}+1$, $V_m(\tilde{n_{ci}}, n_{c,-i}) \leq V_m(n^*_{ci}+1, n_{c,-i})$. This result contradicts the SPE with $\tilde{n_{ci}}$ entrants since those entrants have a profitable deviation not to enter. Thus, only an equilibrium with $n^*_{ci}$ entrants exists.

  I conclude from this part of the proof that in this game, SPE with $n^*_{ci} > 0$ is received iff $V_m(n^*_{ci} + 1, n_{c,-i}) < F \leq V_m(n^*_{ci}, n_{c,-i})$

*Q.E.D*

# B Multiple Equilibria Example

Consider a city with one center ($C = 1$) and two industries ($I = 2$). Let $n_{ci}$ be the number of firms in industry $i$ and $n_{cj}$ the number of firms in industry $j$. We define the (per-firm) variable profits for each combination of $(n_{ci}, n_{cj})$ via the following tables:

Variable Profit for Industry $i$ : $V_i(n_{ci}, n_{cj})$

| | $n_{cj} = 1$ | $n_{cj} = 2$ |
|---|---|---|
| $n_{ci} = 1$ | 9 | 10 |
| $n_{ci} = 2$ | 9 | 6 |
| $n_{ci} = 3$ | 5 | 4 |

Variable Profit for Industry $j$ : $V_j(n_{cj}, n_{ci})$

| | $n_{ci} = 1$ | $n_{ci} = 2$ |
|---|---|---|
| $n_{cj} = 1$ | 10 | 10 |
| $n_{cj} = 2$ | 9 | 7 |
| $n_{cj} = 3$ | 6 | 4 |

Let the fixed entry cost be $F = 8$.

**Equilibrium 1:** $(n_{ci} = 1,\ n_{cj} = 2)$**.**

- **Industry** $i$**:** When $(n_{ci}, n_{cj}) = (1, 2)$,

$$V_i(1, 2) = 10 \ > \ F = 8.$$

  A second entrant in $i$ would yield $(n_{ci}, n_{cj}) = (2, 2)$ with

$$V_i(2, 2) = 6 \ < \ 8,$$

  so another entry to industry $i$ is not profitable.

- **Industry** $j$**:** With $n_{ci} = 1$, having two firms in $j$ yields

$$V_j(2, 1) = 9 \ > \ 8.$$

  Attempting three entrants ($n_{cj} = 3$) gives $V_j(3, 1) = 6 < 8$, so another entry to industry $j$ is not profitable

Hence, $(n_{ci} = 1, n_{cj} = 2)$ is stable for both industries.

**Equilibrium 2:** $(n_{ci} = 2,\ n_{cj} = 1)$**.**

- **Industry** $i$**:** Given $n_{cj} = 1$,

$$V_i(2, 1) = 9 \ > \ 8,$$

  while adding a third entrant ($n_{ci} = 3$) drops profit to $V_i(3, 1) = 5 < 8$. Therefore,another entry to industry $i$ is not profitable.

- **Industry $j$:** With two entrants in $i$, i.e. ($n_{ci} = 2$),

$$V_j(1,2) = 10 \; > \; 8,$$

  but a second firm in $j$ ($n_{cj} = 2$) yields $V_j(2,2) = 7 < 8$. Thus, another entry to industry $j$ is not profitable

As a result, both $(1,2)$ and $(2,1)$ satisfy equilibrium conditions. This example illustrates that a model with several industries affecting each other can support *multiple equilibria*.

# C Additional Estimation Tables

The following tables report the full estimation results for the robustness checks described in Section 5.1.

Table 6: Competitive Effects: 300 Meter Transport Alternative

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -1.039 | -1.055 | -0.273 | -0.089 | -0.734* | -0.162 | -0.901*** | -0.319 | -0.069 | 1.109 |
| | (3.089) | (1.004) | (0.348) | (0.067) | (0.410) | (0.343) | (0.344) | (0.400) | (0.076) | (0.676) |
| Pop 0 ($\beta_1$) | -0.057 | -0.168 | -0.052 | 0.629* | -0.163*** | -0.046 | -0.191*** | -0.267 | 0.238 | 0.069 |
| | (0.162) | (0.141) | (0.115) | (0.340) | (0.058) | (0.246) | (0.037) | (0.212) | (0.272) | (0.067) |
| Pop 1 ($\beta_2$) | -0.033 | 0.020 | 0.124 | -0.119 | 0.038 | -0.172 | 0.014 | 0.039 | -0.188 | 0.023 |
| | (0.074) | (0.023) | (0.152) | (0.092) | (0.026) | (0.391) | (0.015) | (0.033) | (0.200) | (0.022) |
| Pop 2 ($\beta_3$) | 0.029 | 0.006 | -0.048 | 0.126* | -0.014 | 0.134 | 0.012 | -0.011 | 0.120 | 0.004 |
| | (0.076) | (0.011) | (0.055) | (0.072) | (0.012) | (0.286) | (0.009) | (0.018) | (0.136) | (0.008) |
| Bus Stations ($\beta_4$) | -0.002 | -0.004 | 0.008 | -0.033** | -0.004 | 0.008 | -0.007* | -0.001 | -0.014 | 0.005 |
| | (0.008) | (0.005) | (0.014) | (0.017) | (0.005) | (0.021) | (0.004) | (0.008) | (0.021) | (0.005) |
| Road Length ($\beta_5$) | 0.010 | 0.016 | 0.060 | -0.013 | 0.035 | 0.056 | -0.012 | 0.002 | 0.124 | -0.012 |
| | (0.033) | (0.018) | (0.070) | (0.044) | (0.023) | (0.132) | (0.011) | (0.024) | (0.119) | (0.013) |
| $\delta_2$ | 0.887*** | 0.892*** | 0.584*** | 0.373*** | 0.666*** | 0.524 | 0.833*** | 0.757*** | 0.418** | 0.621*** |
| | (0.279) | (0.122) | (0.201) | (0.101) | (0.094) | (0.464) | (0.095) | (0.111) | (0.195) | (0.073) |
| $\delta_3$ | 0.879*** | 0.825*** | 0.353*** | 0.339*** | 0.487*** | 0.620 | 0.641*** | 0.691*** | 0.234** | 0.346*** |
| | (0.321) | (0.202) | (0.135) | (0.095) | (0.102) | (0.497) | (0.104) | (0.124) | (0.115) | (0.056) |
| $\delta_4$ | 0.199 | 0.445* | 0.219** | 0.144*** | 0.356*** | 0.349 | 0.657*** | 0.430*** | 0.240** | 0.248*** |
| | (0.214) | (0.248) | (0.112) | (0.054) | (0.123) | (0.295) | (0.152) | (0.133) | (0.120) | (0.052) |
| $\delta_5$ | 1.002 | 0.255 | 0.613** | 0.177*** | 0.248* | 1.877 | 0.188** | 0.343** | 0.282* | 0.217*** |
| | (1.075) | (0.250) | (0.304) | (0.067) | (0.140) | (3.704) | (0.092) | (0.154) | (0.147) | (0.053) |
| Log-Likelihood | 213.884 | 226.212 | 247.551 | 354.728 | 273.766 | 220.693 | 330.469 | 322.303 | 318.642 | 444.637 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 300 m; `Road_Length` — total length of main roads within 300 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$). No bounds were used except $\delta \geq 0$.

Table 7: Competitive Effects: Socio-Economic Index Alternative

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Socio-Economic Index ($\alpha_1$) | -1.325 | -0.915* | -0.216 | -0.048 | -1.136 | -0.369 | -1.123*** | -0.730 | -0.428 | 2.137 |
| | (2.534) | (0.506) | (0.247) | (0.040) | (0.695) | (0.513) | (0.427) | (0.592) | (1.011) | (1.581) |
| Pop 0 ($\beta_1$) | 0.012 | -0.156 | -0.149 | 1.558 | -0.023 | 0.042 | -0.076* | -0.081 | 0.056 | 0.047 |
| | (0.089) | (0.095) | (0.225) | (1.047) | (0.059) | (0.186) | (0.042) | (0.109) | (0.153) | (0.051) |
| Pop 1 ($\beta_2$) | -0.029 | 0.015 | 0.167 | -0.262 | 0.018 | -0.101 | 0.004 | 0.017 | -0.031 | 0.012 |
| | (0.049) | (0.021) | (0.196) | (0.204) | (0.016) | (0.172) | (0.013) | (0.019) | (0.115) | (0.012) |
| Pop 2 ($\beta_3$) | 0.028 | 0.017 | -0.025 | 0.202 | 0.008 | 0.097 | 0.013* | -0.001 | 0.021 | 0.001 |
| | (0.050) | (0.013) | (0.041) | (0.161) | (0.008) | (0.139) | (0.007) | (0.008) | (0.071) | (0.003) |
| Bus Stations ($\beta_4$) | -0.003 | -0.016* | -0.017 | -0.139* | -0.008 | -0.016 | -0.009* | -0.006 | -0.008 | 0.006 |
| | (0.007) | (0.010) | (0.033) | (0.078) | (0.007) | (0.022) | (0.005) | (0.007) | (0.022) | (0.006) |
| Road Length ($\beta_5$) | -0.012 | 0.013 | 0.071 | 0.332 | -0.030 | -0.019 | -0.050** | -0.020 | 0.058 | -0.019 |
| | (0.028) | (0.023) | (0.117) | (0.249) | (0.024) | (0.057) | (0.024) | (0.028) | (0.109) | (0.018) |
| $\delta_2$ | 0.877*** | 0.945*** | 0.628*** | 0.242** | 0.788*** | 0.727*** | 0.819*** | 0.776*** | 0.645*** | 0.611*** |
| | (0.160) | (0.122) | (0.175) | (0.099) | (0.101) | (0.225) | (0.090) | (0.093) | (0.149) | (0.066) |
| $\delta_3$ | 0.902*** | 0.775*** | 0.358*** | 0.230** | 0.554*** | 0.836*** | 0.741*** | 0.736*** | 0.358*** | 0.359*** |
| | (0.244) | (0.189) | (0.120) | (0.092) | (0.113) | (0.250) | (0.118) | (0.119) | (0.099) | (0.055) |
| $\delta_4$ | 0.145 | 0.416* | 0.215** | 0.101** | 0.405*** | 0.415 | 0.762*** | 0.410*** | 0.347*** | 0.261*** |
| | (0.144) | (0.234) | (0.103) | (0.048) | (0.138) | (0.313) | (0.178) | (0.121) | (0.110) | (0.052) |
| $\delta_5$ | 1.506 | 0.231 | 0.648** | 0.125** | 0.273* | 1.005 | 0.312** | 0.356** | 0.399*** | 0.230*** |
| | (2.310) | (0.226) | (0.294) | (0.059) | (0.155) | (1.241) | (0.151) | (0.156) | (0.154) | (0.055) |
| Log-Likelihood | 215.635 | 224.632 | 253.121 | 351.284 | 270.873 | 220.185 | 316.326 | 319.705 | 319.048 | 449.626 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Socio-Economic Index` — average CBS index of near population; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$). No bounds were used except $\delta \geq 0$.

Table 8: Competitive Effects: First Population Alternative

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -0.750 | -0.725 | -0.183 | -0.016 | -1.040 | -0.085 | -0.656*** | -0.649 | -0.127 | 0.876* |
| | (0.532) | (0.584) | (0.204) | ( NA) | (2.201) | (0.109) | (0.190) | (0.530) | (0.215) | (0.467) |
| Pop 0 ($\beta_1$) | 0.031 | -0.161 | -0.182 | 1.691*** | -0.043 | 0.087 | -0.056 | -0.006 | 0.050 | 0.105 |
| | (0.098) | (0.143) | (0.207) | (0.400) | (0.113) | (0.300) | (0.045) | (0.093) | (0.200) | (0.106) |
| Pop 1 ($\beta_2$) | -0.045 | 0.026 | 0.209 | -0.215 | 0.042 | -0.267 | -0.030 | 0.018 | -0.110 | 0.027 |
| | (0.034) | (0.031) | (0.234) | (0.289) | (0.099) | (0.344) | (0.023) | (0.025) | (0.173) | (0.027) |
| Pop 2 ($\beta_3$) | 0.042 | 0.018 | -0.058 | 0.126 | -0.014 | 0.203 | 0.052*** | 0.003 | 0.071 | 0.008 |
| | (0.029) | (0.022) | (0.076) | (0.199) | (0.040) | (0.251) | (0.018) | (0.015) | (0.109) | (0.016) |
| Pop 3 ($\beta_4$) | 0.003 | 0.001 | 0.016 | 0.165 | 0.011 | 0.076 | -0.015** | -0.002 | 0.008 | -0.002 |
| | (0.007) | (0.006) | (0.025) | ( NA) | (0.023) | (0.097) | (0.006) | (0.005) | (0.032) | (0.005) |
| Bus Stations ($\beta_5$) | -0.005 | -0.016 | -0.015 | -0.201 | -0.006 | -0.031 | -0.017*** | -0.009 | -0.019 | 0.017 |
| | (0.009) | (0.011) | (0.033) | ( NA) | (0.009) | (0.053) | (0.006) | (0.008) | (0.038) | (0.014) |
| Road Length ($\beta_6$) | -0.021 | 0.004 | 0.078 | 0.600 | -0.049 | -0.056 | -0.058** | -0.029 | 0.111 | -0.044 |
| | (0.032) | (0.028) | (0.116) | ( NA) | (0.090) | (0.143) | (0.025) | (0.035) | (0.140) | (0.039) |
| $\delta_2$ | 0.878*** | 0.868*** | 0.571*** | 0.133 | 0.742*** | 0.411 | 0.927*** | 0.744*** | 0.540** | 0.601*** |
| | (0.136) | (0.132) | (0.192) | ( NA) | (0.135) | (0.294) | (0.107) | (0.092) | (0.224) | (0.072) |
| $\delta_3$ | 0.797*** | 0.864*** | 0.339*** | 0.122 | 0.562*** | 0.522 | 0.863*** | 0.721*** | 0.300** | 0.340*** |
| | (0.202) | (0.224) | (0.129) | ( NA) | (0.120) | (0.354) | (0.141) | (0.116) | (0.130) | (0.056) |
| $\delta_4$ | 0.153 | 0.453* | 0.206* | 0.056 | 0.416*** | 0.287 | 0.889*** | 0.440*** | 0.296** | 0.243*** |
| | (0.150) | (0.256) | (0.106) | ( NA) | (0.142) | (0.254) | (0.208) | (0.130) | (0.133) | (0.051) |
| $\delta_5$ | 0.806 | 0.212 | 0.595** | 0.076 | 0.292* | 0.885 | 0.329** | 0.388** | 0.333** | 0.213*** |
| | (0.692) | (0.208) | (0.291) | ( NA) | (0.168) | (1.082) | (0.160) | (0.169) | (0.168) | (0.053) |
| Log-Likelihood | 213.996 | 224.032 | 250.758 | 351.561 | 266.706 | 218.883 | 315.216 | 319.926 | 319.826 | 443.477 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `Pop_0`, `Pop_1`, `Pop_2`, `Pop_3` — populations in own, adjacent, secondary-adjacent and third-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$). No bounds were used except $\delta \geq 0$.

Table 9: Competitive Effects: Second Population Alternative

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -1.308 | -1.761** | -0.423 | -0.123 | -1.830 | -2.704 | -0.090** | -1.245 | -1.611 | 1.261 |
| | (3.095) | (0.895) | (0.835) | (0.092) | (1.683) | (1.962) | (0.041) | (1.512) | (1.979) | (1.149) |
| Pop 0 ($\beta_1$) | -0.073 | -0.077 | -0.091 | 0.454 | -0.031 | -0.048 | -0.074*** | 0.004 | -0.022 | 0.058 |
| | (0.181) | (0.049) | (0.166) | (0.386) | (0.039) | (0.038) | (0.001) | (0.050) | (0.040) | (0.078) |
| Pop 1 ($\beta_2$) | 0.024 | 0.027** | 0.056 | 0.022 | 0.021 | 0.014 | 0.160*** | 0.013 | 0.003 | 0.026 |
| | (0.054) | (0.013) | (0.110) | (0.040) | (0.018) | (0.010) | (0.000) | (0.014) | (0.005) | (0.025) |
| Bus Stations ($\beta_3$) | -0.001 | -0.008 | -0.006 | -0.060* | -0.005 | -0.001 | -0.042 | -0.006 | -0.001 | 0.011 |
| | (0.005) | (0.005) | (0.023) | (0.032) | (0.005) | (0.003) | ( NA) | (0.007) | (0.004) | (0.013) |
| Road Length ($\beta_4$) | 0.012 | 0.008 | 0.043 | 0.223* | -0.014 | 0.006 | -0.025 | -0.017 | 0.022 | -0.028 |
| | (0.030) | (0.013) | (0.102) | (0.133) | (0.016) | (0.010) | ( NA) | (0.023) | (0.028) | (0.035) |
| $\delta_2$ | 0.914*** | 1.022*** | 0.678*** | 0.367*** | 0.774*** | 0.895*** | 0.493*** | 0.751*** | 0.699*** | 0.624*** |
| | (0.177) | (0.121) | (0.214) | (0.075) | (0.098) | (0.107) | (0.052) | (0.086) | (0.081) | (0.079) |
| $\delta_3$ | 0.849*** | 0.773*** | 0.382*** | 0.334*** | 0.599*** | 0.952*** | 0.430*** | 0.681*** | 0.395*** | 0.354*** |
| | (0.232) | (0.186) | (0.141) | (0.071) | (0.121) | (0.213) | (0.068) | (0.107) | (0.079) | (0.061) |
| $\delta_4$ | 0.156 | 0.424* | 0.225** | 0.142*** | 0.375*** | 0.610 | 0.403*** | 0.403*** | 0.370*** | 0.252*** |
| | (0.155) | (0.238) | (0.113) | (0.047) | (0.128) | (0.430) | (0.094) | (0.118) | (0.096) | (0.055) |
| $\delta_5$ | 1.434 | 0.254 | 0.680** | 0.173*** | 0.284* | 0.923 | 0.136** | 0.338** | 0.411*** | 0.220*** |
| | (2.211) | (0.248) | (0.323) | (0.058) | (0.160) | (1.536) | (0.067) | (0.147) | (0.142) | (0.056) |
| Log-Likelihood | 224.442 | 224.765 | 251.410 | 355.915 | 271.071 | 226.281 | 341.596 | 319.626 | 320.165 | 443.425 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `Pop_0`, `Pop_1` — populations in own and adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$). No bounds were used except $\delta \geq 0$.

Table 10: Competitive Effects: FC Entry Effects Alternative

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -0.071 | -0.299*** | -0.122 | -0.069 | -0.010*** | -0.035 | -0.183 | -0.077 | -0.024*** | 0.187 |
| | ( NA) | (0.102) | ( NA) | (0.061) | (0.003) | (0.045) | ( NA) | ( NA) | (0.005) | ( NA) |
| Pop 0 ($\beta_1$) | 0.026 | -0.386 | -0.628 | 1.023 | -0.281 | -0.894 | -0.362*** | 0.015 | 0.012 | 0.268 |
| | (0.998) | (0.244) | ( NA) | (1.001) | ( NA) | (8.787) | (0.075) | (0.700) | (1.409) | (0.290) |
| Pop 1 ($\beta_2$) | -0.739 | 0.059 | 0.255** | -0.194 | 2.295*** | -1.011 | 0.011 | 0.110 | -0.228 | 0.184 |
| | ( NA) | (0.068) | (0.127) | ( NA) | (0.000) | (1.779) | (0.072) | (0.135) | (0.248) | ( NA) |
| Pop 2 ($\beta_3$) | 0.627 | 0.054* | -0.046 | 0.135 | 0.800** | 1.003 | 0.083 | 0.012 | 0.042 | -0.005 |
| | ( NA) | (0.029) | ( NA) | ( NA) | (0.358) | (1.354) | ( NA) | (0.064) | ( NA) | (0.043) |
| Bus Stations ($\beta_4$) | -0.059 | -0.054* | -0.058 | -0.090 | -0.872 | -0.137 | -0.062 | -0.089 | 0.102 | 0.057 |
| | (0.051) | (0.031) | ( NA) | (0.104) | (0.709) | (0.263) | ( NA) | ( NA) | (0.145) | ( NA) |
| Road Length ($\beta_5$) | -0.248** | 0.015 | 0.335 | 0.202 | -3.283 | -0.210 | -0.290 | -0.170 | 1.110 | -0.104 |
| | (0.124) | (0.085) | ( NA) | (0.431) | (2.670) | (1.083) | ( NA) | (0.160) | (0.724) | (0.124) |
| $\gamma_2$ | 0.986*** | 1.019*** | 0.822*** | 0.300*** | 0.838*** | 0.870 | 0.840*** | 0.781*** | 0.733 | 0.630*** |
| | (0.106) | (0.141) | (0.149) | (0.106) | (0.086) | (0.704) | (0.098) | (0.114) | ( NA) | (0.124) |
| $\gamma_3$ | 0.903*** | 0.801*** | 0.198 | 0.219** | 0.626*** | 1.066 | 0.725*** | 0.692*** | 0.400*** | 0.393*** |
| | (0.262) | (0.200) | ( NA) | (0.090) | (0.122) | (0.746) | (0.126) | (0.113) | (0.111) | (0.101) |
| $\gamma_4$ | 0.000 | 0.425 | 0.000 | 0.168 | 0.558*** | 0.000 | 0.672*** | 0.495*** | 0.356*** | 0.282*** |
| | ( NA) | (0.285) | ( NA) | (0.229) | (0.191) | (97.860) | (0.186) | (0.182) | (0.101) | (0.068) |
| $\gamma_5$ | 1.410 | 0.230 | 0.765** | 0.004 | 0.000 | 1.786 | 0.247 | 0.337** | 0.000 | 0.079 |
| | (25.594) | (0.288) | (0.318) | (0.329) | ( NA) | (31.015) | (0.166) | (0.166) | ( NA) | (0.126) |
| $\delta_2$ | 0.000 | 1.310*** | 0.627*** | 0.414** | 0.000 | 0.023 | 0.251** | 0.283* | 0.000 | 0.482*** |
| | (0.042) | (0.427) | (0.209) | (0.163) | (0.014) | (0.246) | (0.111) | (0.150) | ( NA) | (0.166) |
| $\delta_3$ | 0.000 | 0.277* | 0.638 | 0.435* | 0.000 | 0.000 | 0.147** | 0.216** | 0.000 | 0.217* |
| | (0.169) | (0.154) | ( NA) | (0.242) | (0.010) | (0.371) | (0.063) | (0.101) | (0.051) | (0.119) |
| $\delta_4$ | 0.136 | 0.000 | 0.851 | 0.000 | 0.000 | 2.144 | 0.000 | 0.000 | 0.023 | 0.118** |
| | ( NA) | (0.316) | ( NA) | (0.773) | (0.021) | (82.016) | (0.129) | (0.324) | (0.056) | (0.057) |
| $\delta_5$ | 1.011 | 0.000 | 0.000 | 0.571 | 3.870 | 0.000 | 0.000 | 0.000 | 0.987 | 1.310 |
| | (61.996) | (0.567) | (0.335) | (1.266) | ( NA) | (25.933) | (0.078) | (0.102) | ( NA) | ( NA) |
| Log-Likelihood | 211.888 | 222.155 | 248.691 | 350.477 | 251.406 | 216.399 | 313.410 | 318.287 | 317.798 | 442.223 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\gamma_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\gamma_5$) on fixed costs. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$) on variable profit. No bounds were used except $\delta, \gamma \geq 0$.

Table 11: Results 2.0

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -0.833 | -0.052 | -0.137 | -0.102 | -0.606 | -0.429 | -0.742* | 1.084 | -0.913 | 0.017 |
| | (0.872) | (0.488) | (0.592) | (0.088) | (1.248) | (0.372) | (0.395) | (1.087) | (0.586) | (0.081) |
| # Center Categories ($\alpha_2$) | -0.078 | -0.300 | -0.320 | 0.150*** | 1.148*** | 0.175*** | -0.291 | -1.249 | 0.568* | 0.303** |
| | (0.719) | (0.633) | (1.232) | (0.046) | (0.342) | (0.064) | (0.389) | (1.298) | (0.291) | (0.150) |
| Pop 0 ($\beta_1$) | 0.005 | -0.051 | -0.073 | 1.306 | -0.006 | 0.225 | -0.044 | 0.039 | 0.331 | -0.041 |
| | (0.031) | (0.053) | (0.115) | (1.176) | (0.021) | (0.300) | (0.035) | (0.063) | (0.249) | (0.136) |
| Pop 1 ($\beta_2$) | -0.026 | 0.008 | 0.022 | -0.187 | 0.003 | -0.203 | 0.004 | 0.007 | -0.040 | 0.103 |
| | (0.020) | (0.016) | (0.043) | (0.218) | (0.008) | (0.168) | (0.012) | (0.020) | (0.051) | (0.089) |
| Pop 2 ($\beta_3$) | 0.024 | 0.020 | 0.001 | 0.156 | 0.008 | 0.161 | 0.013 | 0.015 | 0.020 | 0.010 |
| | (0.018) | (0.020) | (0.008) | (0.167) | (0.009) | (0.126) | (0.008) | (0.020) | (0.026) | (0.026) |
| Bus Stations ($\beta_4$) | -0.003 | -0.007 | 0.001 | -0.133 | -0.005 | -0.009 | -0.010* | -0.006 | -0.016 | 0.032 |
| | (0.004) | (0.008) | (0.006) | (0.085) | (0.005) | (0.026) | (0.005) | (0.008) | (0.018) | (0.027) |
| Road Length ($\beta_5$) | -0.008 | -0.010 | 0.019 | 0.465 | -0.019 | -0.059 | -0.050** | -0.033 | 0.085 | -0.077 |
| | (0.013) | (0.020) | (0.039) | (0.324) | (0.019) | (0.082) | (0.025) | (0.036) | (0.085) | (0.070) |
| $\delta_2$ | 0.923*** | 0.851*** | 0.738*** | 0.242** | 0.854*** | 0.671*** | 0.860*** | 0.651*** | 0.678*** | 0.568*** |
| | (0.115) | (0.121) | (0.130) | (0.102) | (0.109) | (0.190) | (0.096) | (0.084) | (0.094) | (0.090) |
| $\delta_3$ | 0.948*** | 0.781*** | 0.412*** | 0.226** | 0.510*** | 0.774*** | 0.691*** | 0.614*** | 0.370*** | 0.296*** |
| | (0.230) | (0.193) | (0.112) | (0.093) | (0.104) | (0.230) | (0.112) | (0.101) | (0.079) | (0.062) |
| $\delta_4$ | 0.174 | 0.410* | 0.244** | 0.100** | 0.336*** | 0.476 | 0.741*** | 0.368*** | 0.353*** | 0.205*** |
| | (0.171) | (0.232) | (0.109) | (0.048) | (0.116) | (0.381) | (0.174) | (0.108) | (0.094) | (0.051) |
| $\delta_5$ | 0.791 | 0.246 | 0.794** | 0.122** | 0.260* | 1.312 | 0.279** | 0.309** | 0.395*** | 0.175*** |
| | (0.678) | (0.241) | (0.345) | (0.058) | (0.146) | (1.565) | (0.136) | (0.135) | (0.139) | (0.050) |
| Log-Likelihood | 211.604 | 220.327 | 248.837 | 350.418 | 267.308 | 219.194 | 316.643 | 316.730 | 312.285 | 431.646 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers.

Variables: `Income` — average monthly income per capita; `# Center Categories` — number of unique categories active in the center, excluding the industry's own category; `Pop 0`, `Pop 1`, `Pop 2` — populations in own, adjacent, and secondary-adjacent areas; `Bus Stations` — number of bus stations within 100 m; `Road Length` — total length of main roads within 100 m.

$\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$) on variable profit. No bounds were used except $\delta \geq 0$.

Table 12: Results 2.1

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -0.827 | -0.826 | -0.273 | -0.160 | -0.564 | -0.305* | -0.085** | -0.092 | -0.774 | -0.071 |
| | (0.543) | (0.656) | (0.319) | (0.116) | (0.919) | (0.171) | (0.034) | (1.073) | (0.793) | (0.119) |
| # Center Industries ($\alpha_2$) | -0.012 | -0.148 | 0.708*** | 0.039 | -0.661 | 0.066* | 0.033*** | -0.409 | 0.185*** | 0.198* |
| | (0.121) | (0.265) | (0.034) | (0.027) | (0.550) | (0.037) | (0.012) | (0.448) | (0.071) | (0.107) |
| Pop 0 ($\beta_1$) | 0.010 | -0.050 | -0.149 | 0.477 | -0.005 | 0.654 | -0.197 | 0.021 | 0.166 | -0.085 |
| | (0.080) | (0.055) | (0.193) | (0.831) | (0.017) | (0.500) | (0.204) | (0.037) | (0.235) | (0.109) |
| Pop 1 ($\beta_2$) | -0.047 | 0.008 | 0.135 | -0.028 | 0.005 | -0.377* | 0.044 | 0.016 | -0.036 | 0.068 |
| | (0.032) | (0.015) | (0.178) | (0.152) | (0.007) | (0.203) | (0.123) | (0.016) | (0.079) | (0.073) |
| Pop 2 ($\beta_3$) | 0.044 | 0.015 | -0.021 | 0.061 | 0.005 | 0.279* | 0.170 | -0.002 | 0.029 | 0.019 |
| | (0.029) | (0.014) | (0.038) | (0.109) | (0.004) | (0.146) | ( NA) | (0.006) | (0.053) | (0.019) |
| Bus Stations ($\beta_4$) | -0.005 | -0.010 | -0.014 | -0.090 | -0.003 | 0.002 | -0.044 | -0.003 | -0.024 | 0.015 |
| | (0.008) | (0.011) | (0.031) | (0.073) | (0.003) | (0.040) | ( NA) | (0.004) | (0.026) | (0.018) |
| Road Length ($\beta_5$) | -0.013 | -0.003 | 0.086 | 0.764* | -0.015 | -0.098 | -0.176*** | -0.022 | 0.134 | -0.043 |
| | (0.025) | (0.016) | (0.127) | (0.439) | (0.010) | (0.139) | (0.053) | (0.021) | (0.120) | (0.047) |
| $\delta_2$ | 0.916*** | 0.914*** | 0.634*** | 0.238** | 0.849*** | 0.559*** | 0.408*** | 0.757*** | 0.641*** | 0.621*** |
| | (0.132) | (0.117) | (0.176) | (0.097) | (0.105) | (0.162) | (0.072) | (0.088) | (0.129) | (0.101) |
| $\delta_3$ | 0.814*** | 0.869*** | 0.364*** | 0.211** | 0.550*** | 0.688*** | 0.364*** | 0.703*** | 0.350*** | 0.326*** |
| | (0.202) | (0.211) | (0.122) | (0.085) | (0.112) | (0.217) | (0.082) | (0.109) | (0.091) | (0.069) |
| $\delta_4$ | 0.113 | 0.452* | 0.221** | 0.093** | 0.385*** | 0.513 | 0.387*** | 0.415*** | 0.329*** | 0.226*** |
| | (0.112) | (0.254) | (0.106) | (0.044) | (0.131) | (0.469) | (0.110) | (0.120) | (0.102) | (0.058) |
| $\delta_5$ | 1.038 | 0.250 | 0.665** | 0.112** | 0.295* | 1.076 | 0.168** | 0.349** | 0.367*** | 0.194*** |
| | (1.039) | (0.245) | (0.301) | (0.053) | (0.165) | (1.194) | (0.085) | (0.151) | (0.139) | (0.056) |
| Log-Likelihood | 213.903 | 223.326 | 251.126 | 341.757 | 266.939 | 212.633 | 331.088 | 318.318 | 309.481 | 424.695 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `# Center Industries` - number of unique industries active in the center, without the industry inspected; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$) on variable profit. No bounds were used except $\delta \geq 0$.

Table 13: Results 2.2

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -1.014 | -0.650 | -0.352 | -0.217 | -0.159 | -0.226* | -0.193 | 0.068 | -0.777 | 0.212 |
| | (0.617) | (0.445) | (0.425) | (0.139) | (0.527) | (0.133) | ( NA) | (1.029) | (0.928) | (0.179) |
| # Center Firms ($\alpha_2$) | -0.062 | -0.059 | 0.229*** | 0.027* | 0.339 | 0.053*** | -0.170 | -0.369 | 0.185** | 0.146 |
| | (0.100) | (0.080) | (0.071) | (0.016) | (0.247) | (0.015) | ( NA) | (0.424) | (0.093) | (0.537) |
| Pop 0 ($\beta_1$) | 0.006 | -0.069 | -0.123 | 0.319 | -0.014 | 0.263 | -0.059*** | 0.014 | 0.120 | 0.002 |
| | (0.050) | (0.060) | (0.130) | (0.478) | (0.024) | (0.322) | (0.016) | (0.029) | (0.151) | (0.052) |
| Pop 1 ($\beta_2$) | -0.034* | 0.012 | 0.062 | 0.002 | 0.008 | -0.347 | 0.010 | 0.014 | -0.021 | 0.048 |
| | (0.020) | (0.019) | (0.100) | (0.092) | (0.010) | ( NA) | (0.013) | (0.017) | (0.052) | (0.056) |
| Pop 2 ($\beta_3$) | 0.031* | 0.019 | -0.007 | 0.010 | 0.008 | 0.260*** | 0.008 | -0.001 | 0.016 | 0.007 |
| | (0.017) | (0.015) | (0.020) | (0.056) | (0.006) | (0.053) | ( NA) | (0.004) | (0.033) | (0.011) |
| Bus Stations ($\beta_4$) | -0.004 | -0.012 | -0.001 | -0.078* | -0.005 | 0.013 | -0.005 | -0.002 | -0.006 | 0.006 |
| | (0.005) | (0.010) | (0.015) | (0.047) | (0.004) | (0.040) | ( NA) | (0.003) | (0.015) | (0.009) |
| Road Length ($\beta_5$) | -0.008 | -0.007 | 0.034 | 0.555** | -0.018 | -0.047 | -0.041 | -0.016 | 0.066 | -0.033 |
| | (0.016) | (0.021) | (0.069) | (0.265) | (0.013) | (0.128) | ( NA) | (0.019) | (0.080) | (0.035) |
| $\delta_2$ | 0.923*** | 0.900*** | 0.695*** | 0.306*** | 0.793*** | 0.546*** | 0.812*** | 0.710*** | 0.650*** | 0.653*** |
| | (0.117) | (0.118) | (0.149) | (0.085) | (0.100) | (0.173) | (0.089) | (0.082) | (0.115) | (0.095) |
| $\delta_3$ | 0.896*** | 0.818*** | 0.401*** | 0.272*** | 0.604*** | 0.638*** | 0.642*** | 0.652*** | 0.363*** | 0.351*** |
| | (0.217) | (0.201) | (0.117) | (0.077) | (0.122) | (0.212) | (0.100) | (0.102) | (0.087) | (0.068) |
| $\delta_4$ | 0.124 | 0.421* | 0.231** | 0.119*** | 0.413*** | 0.982 | 0.613*** | 0.391*** | 0.350*** | 0.245*** |
| | (0.123) | (0.238) | (0.105) | (0.045) | (0.141) | ( NA) | (0.139) | (0.113) | (0.101) | (0.059) |
| $\delta_5$ | 1.033 | 0.235 | 0.754** | 0.144*** | 0.263* | 0.920 | 0.275** | 0.340** | 0.379*** | 0.214*** |
| | (1.045) | (0.230) | (0.335) | (0.054) | (0.148) | (0.799) | (0.131) | (0.147) | (0.137) | (0.058) |
| Log-Likelihood | 213.169 | 223.303 | 250.959 | 344.977 | 263.970 | 217.672 | 320.305 | 317.249 | 315.200 | 431.660 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `# Center Firms` - number of firms active in the center, without the firms in the industry; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$) on variable profit. No bounds were used except $\delta \geq 0$.

Table 14: Results 2.3

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -0.850 | -0.873* | -0.286 | -0.054 | -1.311 | -0.369 | -1.134** | -1.008 | -0.473* | 0.326* |
| | (0.774) | (0.502) | (0.307) | (0.044) | (0.893) | (0.556) | (0.442) | (0.967) | (0.262) | (0.187) |
| # Center Categories > 5 ($\alpha_2$) | -0.561 | -0.247 | 0.660*** | 0.281** | 0.378 | 0.629*** | -0.281 | -1.555 | 2.110** | 1.345** |
| | (1.162) | (1.387) | (0.240) | (0.130) | (2.254) | (0.239) | (1.221) | ( NA) | (0.991) | (0.685) |
| Pop 0 ($\beta_1$) | -0.007 | -0.082 | -0.110 | 1.485 | -0.023 | -0.004 | -0.048 | 0.013 | 0.278 | 0.075 |
| | (0.075) | (0.075) | (0.236) | (0.973) | (0.050) | (0.173) | (0.037) | ( NA) | (0.203) | (0.130) |
| Pop 1 ($\beta_2$) | -0.042 | 0.015 | 0.170 | -0.249 | 0.011 | -0.112 | 0.004 | 0.013 | -0.057 | 0.090 |
| | (0.030) | (0.021) | (0.190) | (0.193) | (0.018) | (0.202) | (0.012) | ( NA) | (0.053) | (0.072) |
| Pop 2 ($\beta_3$) | 0.041 | 0.018 | -0.029 | 0.190 | 0.010 | 0.108 | 0.012* | 0.000 | 0.026 | -0.001 |
| | (0.030) | (0.013) | (0.040) | (0.147) | (0.008) | (0.167) | (0.007) | ( NA) | (0.026) | (0.019) |
| Bus Stations ($\beta_4$) | -0.004 | -0.015* | -0.027 | -0.132* | -0.007 | -0.006 | -0.010** | -0.006 | 0.015 | 0.021 |
| | (0.008) | (0.009) | (0.035) | (0.071) | (0.006) | (0.021) | (0.005) | ( NA) | (0.020) | (0.018) |
| Road Length ($\beta_5$) | -0.012 | -0.002 | 0.129 | 0.320 | -0.028 | -0.030 | -0.045** | -0.019 | 0.046 | -0.061 |
| | (0.025) | (0.023) | (0.146) | (0.240) | (0.024) | (0.067) | (0.021) | (0.022) | (0.068) | (0.055) |
| $\delta_2$ | 0.881*** | 0.910*** | 0.605*** | 0.250*** | 0.775*** | 0.694*** | 0.815*** | 0.735 | 0.636*** | 0.561*** |
| | (0.130) | (0.121) | (0.174) | (0.094) | (0.103) | (0.268) | (0.090) | ( NA) | (0.092) | (0.083) |
| $\delta_3$ | 0.874*** | 0.854*** | 0.353*** | 0.237*** | 0.586*** | 0.811*** | 0.718*** | 0.681*** | 0.354*** | 0.308*** |
| | (0.221) | (0.213) | (0.120) | (0.089) | (0.120) | (0.291) | (0.114) | (0.077) | (0.077) | (0.060) |
| $\delta_4$ | 0.127 | 0.441* | 0.219** | 0.104** | 0.407*** | 0.418 | 0.659*** | 0.381*** | 0.350*** | 0.219*** |
| | (0.126) | (0.249) | (0.105) | (0.047) | (0.139) | (0.320) | (0.154) | (0.101) | (0.093) | (0.052) |
| $\delta_5$ | 0.976 | 0.223 | 0.646** | 0.129** | 0.265* | 1.192 | 0.214** | 0.334** | 0.410*** | 0.191*** |
| | (0.947) | (0.219) | (0.293) | (0.058) | (0.150) | (1.568) | (0.105) | (0.134) | (0.143) | (0.052) |
| Log-Likelihood | 213.301 | 223.405 | 248.983 | 352.153 | 269.426 | 219.911 | 315.944 | 319.707 | 312.376 | 437.114 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `# Center Categories > 5` - a dummy variable taking 1 if there are more than five active categories in the center, and 0 else; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$) on variable profit. No bounds were used except $\delta \geq 0$.

Table 15: Results 2.4

| variable | Bakery | Beauty Salon | Cafe | Clothing | Contractor | Electronics | Grocery | Hair Care | Home Goods | Restaurant |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($\alpha_1$) | -0.950 | -1.229** | -0.294 | -0.107 | -1.158 | -0.210 | -0.071* | 0.016 | -0.230 | -0.022 |
| | (0.856) | (0.531) | (0.319) | (0.100) | (1.146) | (0.195) | (0.040) | (1.225) | (0.179) | (0.203) |
| # Center Chosen Industries ($\alpha_2$) | 0.590* | 0.423* | 0.202* | 0.184** | 1.722 | 0.195* | 0.079** | 0.911* | 0.189* | 1.243* |
| | (0.338) | (0.243) | (0.121) | (0.077) | (1.084) | (0.101) | (0.040) | (0.491) | (0.108) | (0.654) |
| Pop 0 ($\beta_1$) | -0.260 | -0.209** | -0.104 | 0.056 | -0.006 | 0.555 | -0.309 | -0.046 | 0.245 | -0.043 |
| | (0.237) | (0.104) | (0.261) | (0.813) | (0.015) | (0.606) | (0.349) | (0.108) | (0.394) | (0.051) |
| Pop 1 ($\beta_2$) | -0.014 | 0.020 | 0.178 | -0.095 | 0.006 | -0.390 | 0.152 | -0.031 | -0.168 | 0.015 |
| | (0.034) | (0.022) | (0.207) | (0.244) | (0.006) | (0.407) | (0.193) | (0.065) | (0.193) | (0.022) |
| Pop 2 ($\beta_3$) | 0.036 | 0.016 | -0.027 | 0.164 | 0.004 | 0.326 | 0.264 | 0.010 | 0.150 | 0.014 |
| | (0.034) | (0.011) | (0.044) | (0.221) | (0.004) | (0.314) | (0.161) | (0.029) | (0.145) | (0.013) |
| Bus Stations ($\beta_4$) | -0.010 | -0.018** | -0.027 | -0.115 | -0.003 | 0.007 | -0.068 | 0.004 | -0.017 | 0.003 |
| | (0.015) | (0.009) | (0.039) | (0.108) | (0.002) | (0.052) | (0.047) | (0.008) | (0.044) | (0.006) |
| Road Length ($\beta_5$) | 0.098 | 0.025 | 0.109 | 0.911 | -0.015 | -0.051 | -0.211* | 0.036 | 0.171 | -0.007 |
| | (0.095) | (0.028) | (0.145) | (0.658) | (0.010) | (0.181) | (0.112) | (0.073) | (0.173) | (0.016) |
| $\delta_2$ | 0.881*** | 0.971*** | 0.616*** | 0.199* | 0.849*** | 0.482* | 0.311*** | 0.767*** | 0.475*** | 0.712*** |
| | (0.175) | (0.121) | (0.176) | (0.115) | (0.105) | (0.250) | (0.105) | (0.084) | (0.161) | (0.101) |
| $\delta_3$ | 0.785*** | 0.886*** | 0.352*** | 0.169* | 0.529*** | 0.568** | 0.268*** | 0.673*** | 0.255*** | 0.371*** |
| | (0.220) | (0.220) | (0.119) | (0.097) | (0.108) | (0.270) | (0.095) | (0.112) | (0.095) | (0.069) |
| $\delta_4$ | 0.151 | 0.489* | 0.211** | 0.072 | 0.381*** | 0.312 | 0.280*** | 0.383*** | 0.237** | 0.260*** |
| | (0.150) | (0.275) | (0.102) | (0.045) | (0.130) | (0.258) | (0.109) | (0.116) | (0.095) | (0.060) |
| $\delta_5$ | 1.215 | 0.254 | 0.637** | 0.088 | 0.322* | 1.236 | 0.123* | 0.311** | 0.258** | 0.228*** |
| | (1.447) | (0.248) | (0.291) | (0.054) | (0.179) | (1.757) | (0.071) | (0.139) | (0.118) | (0.060) |
| Log-Likelihood | 207.172 | 222.053 | 249.984 | 338.221 | 265.421 | 209.314 | 324.166 | 317.946 | 304.744 | 425.136 |

*Notes:* $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Estimation by maximum likelihood (MLE), with standard errors in parentheses (Hessian-based), based on $N = 267$ centers. Variables: `Income` — average monthly income per capita; `# Center Chosen Industries -` number of unique industries active in the center, out of the ten industries of choice, without the industry inspected; `Pop_0`, `Pop_1`, `Pop_2` — populations in own, adjacent, and secondary-adjacent areas; `Bus_Stations` — number of bus stations within 100 m; `Road_Length` — total length of main roads within 100 m. $\delta_k$ — effect of the $k$-th entrant ($n_{ic} > 5$ grouped into $\delta_5$) on variable profit. No bounds were used except $\delta \geq 0$.